THE PREDICTABILITY OF DEMOGRAPHIC AND INSTITUTIONAL FACTORS ON THE PERSISTENCE AND ACADEMIC SUCCESS OF INTERNATIONAL STUDENTS

DISSERTATION

Presented in Partial Fulfilment of the Requirements for

the Degree Doctor of Education in the Graduate School

of Texas Southern University

By

Angela C. Offurum, A.A.S., B.Eng., MHCA

Texas Southern University

2019

Approved By

Dr. Danita Bailey-Samples

_____

Chairperson, Dissertation Committee

Dr. Gregory Maddox

_____

Dean, The Graduate School

ProQuest Number: 27548091

All rights reserved

INFORMATION TO ALL USERS
The quality of this reproduction is dependent on the quality of the copy submitted.

In the unlikely event that the author did not send a complete manuscript
and there are missing pages, these will be noted. Also, if material had to be removed,
a note will indicate the deletion.



ProQuest 27548091

Published by ProQuest LLC (2020). Copyright of the Dissertation is held by the Author.

All Rights Reserved.
This work is protected against unauthorized copying under Title 17, United States Code
Microform Edition © ProQuest LLC.

ProQuest LLC
789 East Eisenhower Parkway
P.O. Box 1346
Ann Arbor, MI 48106 - 1346

Approved By

Dr. Danita Bailey-Samples	November 13, 2019
Chairperson, Dissertation Committee	Date
Dr. Lillian B. Poats	November 15, 2019
Committee Member	Date
Dr. Ronnie Davis	November 15, 2019
Committee Member	Date
Dr. Ronald Samples	November 13, 2019
Committee Member	Date

©COPYRIGHT Angela C. Offurum 2019

All Rights Reserved

THE PREDICTABILITY OF DEMOGRAPHIC AND INSTITUTIONAL FACTORS ON THE PERSISTENCE AND ACADEMIC SUCCESS OF INTERNATIONAL STUDENTS

By

Angela C. Offurum, Ed.D.

Texas Southern University, 2019

Professor Danita Bailey-Samples, Advisor

International students are faced with challenges within institutions that are driven by several factors, among which are demographic factors (*gender* and *ethnicity*) and institutional factors (*campus interactions, school climate, support services*, and *social activities*). Therefore, investigating the predictive power of these factors on the *persistence* and *academic success* of these students was the purpose of the research.

Methodology

The *Persistence* and *Academic Success Inventory* (PASI) questionnaire was administered to a representative sample of international students enrolled in one of the Historically Black Colleges and Universities (HBCUs) in the southern part of the United States. Demographic information on gender and ethnicity was collected. The five-point *Likert* scale, ranging from "*Strongly Disagree"* to "*Strongly Agree,*" and the *Rating* scale, from "*Never"* to "*Always,*" and hours-per-week from "*Less than 1"* to "*More than 15,*" were used to measure student response pertaining to institutional factors. *Logistic* and *Multiple Regression* were used for data analyses, after responses were computed using

the SPSS software. The hypotheses for *persistence* and *academic success* were tested for significance.

Findings

The following major findings were identified:

1. There was no statistically significant relationship between gender, ethnicity, and both *persistence* and *academic success* among the international students.
2. Support services and social activities were significant predictors of *persistence*.
3. No linear relationship was found to exist between gender, ethnicity, and *academic success*.
4. A linear relationship was found to exist between the institutional factors and *academic success*.
5. Campus interactions, support services, and social activities were independent predictors of *academic success*.
6. School climate was not a predictor of both *persistence* and *academic success* among the international students.

Recommendations

1. Investigate marital status, family makeup, sponsorship type, and cross-cultural differences.
2. Research the predictive power of on-campus jobs, scholarships, and institutional assistance.

3. Explore best methods for school administrators to communicate with a culturally diversified student group, producing positive campus experiences, and thus more enrollment referrals to the institution.

Conclusions

Gender and ethnicity were not independent predictors of *persistence* and *academic success* among international students. Thus, international students face challenges no matter their gender or ethnicity. As institutional factors, support services and social activities independently were significant predictors of *persistence* and *academic success.* Campus interactions independently predicted *academic success*, while *school climate* predicted neither *persistence* nor *academic success* among international students.

KEY WORDS: International Students, Foreign Students, Demographic Factors, Institutional Factors, Persistence, Academic Success, Retention, Gender, Ethnicity, Campus Interactions, School Climate, Support Services, Social Activities.

TABLE OF CONTENTS

	Page
LIST OF TABLES ……..	iv
LIST OF FIGURES ……	vi
VITA ……..	vii
ACKNOWLEDGEMENTS ……	viii
DEDICATION ……	ix
CHAPTER	
1. INTRODUCTION ……...	1
2. LITERATURE REVIEW ……	20
3. METHODOLOGY ……..	41
4. DATA RESULTS AND ANALYSES ……	63
5. SUMMARY, RECOMMENDATIONS, AND CONCLUSIONS …….	82
REFERENCES ……	99

LIST OF TABLES

Table		Page
1.	Frequency Distribution of Study Participants by Gender …….	65
2.	Frequency Distribution of Study Participants by Ethnicity ……..	66
3.	Mean and Standard Deviation Results for the Independent and Dependent Variables …….	67
4.	Correlation Results between Independent and Dependent Variables in the Regression Models ……..	69
5.	Overall Model Fit Results Regarding the Relationship between Demographic Factors and Persistence ……..	71
6.	Classification Table Results Regarding Demographic Factors and Persistence …….	71
7.	Regression Coefficients Regarding the Relationship between Gender, Ethnicity, and Persistence ……...	73
8.	Overall Model Fit Results Regarding the Relationship between Institutional Factors and Persistence …….	74
9.	Classification Table Results Regarding Institutional Factors and Persistence …….	75
10.	Regression Coefficients Regarding the Relationship between Institutional Factors and Persistence …….	76
11.	Regression Coefficients Regarding the Relationship between Gender, Ethnicity and Academic Success …….....	77

Table Page

12. Regression Coefficients Regarding the Relationship between Institutional Factors and Academic Success ……. 79

13. Summary Table for the Hypotheses Tested in the Study …….. 81

LIST OF FIGURES

Figure Page

1. Predictive Power of Demographic and Institutional Factors on the Persistence and Academic Success of International Students ……..… 44
2. Independent and Dependent Variables in the Study ……. 54
3. Predictive Power of Demographic and Institutional Factors on the Persistence of International Students …… 57
4. Predictive Power of Demographic and Institutional Factors on the Academic Success of International Students …… 58

VITA

1990……	B.Eng., Federal University of Technology Owerri, Imo State, Nigeria
1992-1997……	Senior Technical Officer Electronic Systems Ltd. Lagos, Nigeria
1997-2004……	Senior Banking Officer Diamond Bank PLC. Lagos, Nigeria
2004-2008……	Television Production Intern Montgomery College, Rockville, Maryland
2007-2008……	Student Intern Voice of America, Washington, DC
2008……..…...	Media Development Intern Department of Environmental Protection, Rockville, Maryland
2009……..…	A.A.S., Montgomery College, Rockville, Maryland
2009-2010 ……... 2012-2019	Graduate Assistant Administrator College of Pharmacy & Health Sciences, Texas Southern University, Houston, Texas
2010-2011……	Volunteer Administrative Intern Bellfort Wellness Center Inc., Houston, Texas
2011…….	MHCA, Texas Southern University, Houston, Texas
2011-2012……	Business Office Specialist Riverside General Hospital, Houston, Texas
Major Field……...	Educational Administration

ACKNOWLEDGEMENTS

My passion for this dissertation topic stemmed from the challenges that I see students facing when they leave their home country to further their studies. As a Student Services staff member, I see these challenges and concerns as these students strive towards academic success. Thus, I thank every student who participated in this study. There would have been no project without your responses to the questionnaire.

Sincere gratitude goes to my dissertation team. Dr. Danita Bailey-Samples, as the committee chair and my advisor, painstakingly guided me throughout the process and helped me meet deadlines. Dr. Lillian Poats gave me great insight on the details and layout of my work. Dr. Ronnie Davis encouraged me with statistical challenges because he felt I could handle them. Dr. Ronald Samples reviewed my work from a different perspective of sentence structure. Thank you.

My appreciation goes to all the faculty and staff who have been there for me at some point or another, especially during the dissertation process. Their help, advice, and guidance were very well appreciated. For those who could not officially be on my thesis committee, thank you for your feedback and recommendations. They helped me immensely during this study.

A special thank you goes to all those in the College of Education, College of Pharmacy and Health Sciences, and other departments and schools within and outside Texas Southern University who have encouraged me throughout this expedition. I also thank family, friends, and others who truly wish the best for me in life. Finally, I say, "Thank you and God bless," to everyone who has touched my life in some way.

DEDICATION

This work is dedicated first to God, who is the author and completer of everything pertaining to me, then to my parents and my brothers and sisters. Collectively, they gave me a reason not to give up on my academic journey, the quest for knowledge, and the joy of living. Having them in my life built resilience and perseverance in me. Without God, I would not have my today, and my tomorrow may not be sure. As for my family, their love will always be there. Thank you.

CHAPTER 1

INTRODUCTION

The education industry is very dynamic and the daily processes of academic pursuit for students are very complex. College students face numerous challenges as they persist through school to reach their desired goals and achieve *academic success*. The international arena is no exception as individuals come from all over the world to the United States for educational purposes. They are known as international students. These international students are persons who have left their home country or country of birth to pursue or further their education in another country. According to Rebolledo-Gómez and Ranchin (2013), international students are those who have crossed borders for the purpose of study. The UNESCO Institute for Statistics (UIS), the Organization for Economic Co-operation and Development (OECD), and the Statistical Office of the European Union (EUROSTAT) also provide their definition of an international student. Rebolledo-Gómez and Ranchin (2013) state that the UIS, the OECD, and EUROSTAT define international students as those who are not residents of their country of study or those who received their prior education in another country. Furthermore, Clark (2009) states that the UIS's *Global Education Digest* sees an international student as one who has left his or her country or territory of origin and moved to another country or territory specifically to study.

Several studies have investigated factors that impact students in general and their academic progress. Sanchez and Smith (2017) examined factors such as gender, grade

point average (GPA), credits earned, type and amount of financial aid, *persistence*, and graduation status. Ambrósio, Marques, Santos, and Doutor (2017) reviewed institutional support. Other studies evaluated *student success*, *persistence*, or dropout, as well as students' experiences without referring to the international student population. However, studies focusing jointly on the *persistence,* and *academic success* of international students are very limited (Mamiseishvili, 2012). Perry, Lausch, Weatherford, Goeken, and Aemendares (2017) stated that international students are a growing segment of the American postsecondary education system, and currently account for approximately 4% of the total student population. The authors also note that international students bring significant revenue to the nation. However, leaving one's country of origin, changing environment from one's home country to adapt elsewhere, while preparing for *academic success*, is full of challenges.

Educators have noted these challenges as they constantly encounter factors that impact student *persistence* through school and ultimately *academic success*. In every level of education, there is a growing need to understand these phenomena. Initially, studies did not investigate the issue of student *persistence*. Demetriou and Schmitz-Sciborski (2011) remarked that the earliest studies of undergraduate retention in the United States occurred in the 1930s and was referred to as student mortality. The authors also noted that in the 1960s, there were multiple publications which started the national dialogue on undergraduate retention. According to Demetriou and Schmitz-Sciborski (2011), Tinto's 1975 landmark publication marked the actual start of the phenomenon of student retention, with more recent versions including a motivational component. Tinto

(2017) states that *persistence* is another way to describe motivation, noting that a student should have that desire to persist towards degree completion in order to put ample effort in the process. Demetriou and Schmitz-Sciborski (2011) acknowledged that these motivational orientations and optimism have now been used to gain understanding into college student *persistence* and retention.

Various scholars have researched the phenomenon of *academic success* on many levels and found it to be interchangeable with *student success* and *academic achievement*. York, Gibson, and Rankin (2015) concur. These authors conducted a comprehensive literature search to help define and measure this phenomenon. They stated that their review of literature revealed that the terms "*student success"* and "*academic success"* were used interchangeably. Based on their findings, York et al. (2015) define *academic success* as inclusive of academic achievement, attaining learning objectives, acquiring desired skills and competencies, satisfaction, *persistence*, and post-college performance. Their results also indicated that grades and GPA were the most commonly used measures of *academic success*.

Steinmayr, Meißner, Weidinger, and Wirthwein (2014) held that academic achievement is very wide-ranging, covering a broad variety of educational outcomes, and its definition is based on the indicators used to measure it. The authors declared that academic achievement included criteria such as grades, achievement test scores, degrees, and certificates. Steinmayr et al. (2014) emphasized the importance of academic achievement in developed societies. Measured by GPA, or the Scholastic Assessment Test (SAT), it determined whether a student would have the opportunity to continue his

or her education, such as attend a university (Steinmayr et al., 2014). Likewise, research by DeFreitas (2012) reported academic performance as related to academic achievement and measured using GPA. Thus, SAT scores were applied.

Persisting in college and succeeding in a chosen major can be daunting to any student. The situation is magnified for the foreign student who also must deal with environmental and social changes. It is thus important to examine the various factors international students encounter, noting the challenges that may appear as they persist through school and gain *academic success*. In light of this, it is vital to study the growing phenomena of *persistence* and *academic success* in the institutions of higher learning in the country. Thus, this study investigated selected demographic and institutional factors, and their power to predict the *persistence* and *academic success* of international students.

Statement of the Problem

Research indicates that international students encounter a unique set of challenges when it comes to academic and social integration, in addition to campus expectations (Schulmann & Choudaha, 2014). Fischer (2014) stated that earlier research on international students investigated their retention and *academic success* within individual campuses or university systems. The author reported research across the nation, where concerns loomed in various educational settings regarding this population of students and the rate at which they leave institutions without earning a degree. Fischer (2014) referred to a study conducted by the World Education Service (WES) and made public by the Association of International Educators (NAFSA). According to Fischer (2014), the study

revealed that educators associated foreign undergraduates' decisions to drop out of school or transfer to another one with a variety of reasons. Such reasons, they stated, included finances, academics, and English-language problems, as well as the desire to attend an institution that is a "better fit."

Noting these reasons for student dropout, the purpose of this study was to examine the predictive power of selected demographic and institutional factors on the *persistence* and *academic success* of international students. More specifically, this study was concerned with the predictive power of demographic factors (*gender* and *ethnicity*) and institutional factors (*campus interactions, school climate, support services*, and *social activities*) on the *persistence* and *academic success* of international students. In the context of this study, demographic factors were categorized as gender and ethnicity. Regarding gender, students were either male or female. Ethnicity referred to the ethnic groups that they fell under. As international students coming from different ethnic groups, the nature of the diversity may have predictive power on their *persistence* (accomplished by their return to the same school the following semester) and *academic success*, as determined by GPA.

International students interact with various individuals while on the school campus. The predictive power of these interactions on student *persistence* and *academic success* was examined. Students' perception of the campus environment and their school climate experiences were sought. Furthermore, support services on campus, as well as the social activities available to these students, were examined to determine whether they

have predictive power on student *persistence* and *academic success*. Answers to the following questions were sought in the study:

1) Do demographic factors (gender and ethnicity) have any predictive power with regard to persistence among international students?
2) Do institutional factors (campus interactions, school climate, support services, and social activities) have any predictive power with regard to persistence among international students?
3) Do demographic factors (gender and ethnicity) have any predictive power with regard to academic success among international students?
4) Do institutional factors (campus interactions, school climate, support services, and social activities) have any predictive power with regard to academic success among international students?

With the notion that international students will remain a significant topic in the educational system, researchers have started investigating this population. They noted the fact that these students are a benefit to the industry and the nation. Perry et al. (2017) acknowledged international students in terms of enrollment and cultural diversity, stating that, in addition to economic benefits, international students contribute to America's understanding of other cultures and practices. With the changes in the educational system and the increase in globalization (Perry et al., 2017), it is important to recognize this group of individuals who are in the middle of the issue. Educators must thus understand that they too may step out into the world and face challenges away from home.

Significance of the Study

The desire for achievement makes individuals become creative. Students who want to excel academically stop at nothing to do so. Challenges may ensue, but determination is the key. This is *persistence.* As mentioned earlier, Tinto (2017) states that *persistence* is the quality that allows someone to continue in pursuit of a goal even when challenges arise. Unfortunately, the plight of the international student is heightened by rules and regulations guiding their activities while pursuing their education in that foreign country. Thus, the *persistence* of international students is an important research phenomenon to consider.

Despite facing challenges while studying away from their country of origin, international students can achieve *academic success.* According to Kupczynski, Brown, Holland, and Uriegas (2014), many specific characteristics of individual students themselves, such as gender, age, race, and learning style, have been noted as factors affecting *student success* in the online classroom. The physical classroom also contributed to *student success*. In this study, particular attention was given to gender, ethnicity, campus interactions, school climate, support services, and social activities as it related to international students. It was therefore necessary to determine the predictive power of these selected factors on the *persistence* and *academic success* of international students, and to ascertain the ultimate impact on the education industry.

When away from their home country, international students encounter varying situations. How these issues are addressed makes a difference in the students' overall perception of the institution. Therefore, a study of this nature is significant in the

education industry with implications on the students' learning experiences in the country. Diversity in methods of teaching, learning, and collaboration can bring about positive experiences which may possibly lead to referrals to the institution. This study sheds light on selected demographic and institutional factors, and how they predict the student's learning process. It also reflected how these selected factors affected students' return to that particular educational institution. This study is thus essential not only to the education industry, but also to the various professional programs, as positive influences are likely to translate into improved academic outcomes for the students and recognition of the quality of that academic institution. This could, again, bring more referrals to the institution, and even possible funding.

This study, particularly unique, is also significant for policy makers. Results and outcomes can provide policy guidelines for best practices in international student recruitment, classroom instruction, and examination preparation, as well as student community involvement. Ultimately, these practices could lead to students' *persistence* and *academic success* in their chosen academic programs, as well as the desire to return to the institution and the host country. For the academic institution, this could mean increased enrollment.

Theoretical Framework

Several theories and pertinent theoretical backgrounds were used to explain these phenomena of student *persistence* and *academic success* under study. This section thus discusses the different theories on which the framework for this dissertation was built.

They include the works of Astin - *Student Involvement*, Tinto - *Student Departure* (Integration Model), Bean - *Student Attrition,* Bandura - *Social Learning Theory*, Walberg - *Theory of Educational Productivity*, and Edgar - *Learning Theories* (Behaviorism).

Student *persistence* became the construct used in several studies which began when authors investigated the issue of student retention, as well as reasons for departure. According to Demetriou and Schmitz-Sciborski (2011), the earliest studies of undergraduate retention appeared in the 1930s. The authors noted a 1938 study led by John McNeely, where institutional data were collected and examined. Over the years, studies have interchanged the concept of *student retention* with *student attrition*, and *student persistence,* based on the angle of interpretation. For instance, Cabrera, Nora, and Castaneda (1993) explain their own studies of college student *persistence* based on previous studies of *student integration* and *student attrition*. Inasmuch as the construct of *academic success* has been used most frequently and interchangeably with *student success* and *academic achievement*, York et al. (2015) state the importance of knowing that it has become a catchall phrase encompassing numerous student outcomes measures.

Astin's theory of student involvement and Tinto's theory of student departure are well researched and cited in studies regarding student *persistence*. Astin (1999) stated that the theory of student involvement had its origin in a longitudinal study of college dropouts. The study endeavored to identify factors in the college environment that significantly affected student's *persistence* in college. According to Astin (1999, 1984), student involvement referred to the amount of physical and psychological energy that the

student devoted to the academic experience. The individual's actions, that is, his or her behavior, define and identify involvement which, in certain respects, resembled the behavioral manifestation of *motivation* (Astin, 1999, 1984).

Another aspect of Astin's theory of student involvement was viewed through the Input-Environment-Output (I-E-O) Model. According to York et al. (2015), dominant definitions of *academic success* were conceptually evaluated using Astin's I-E-O model. Reviewing Astin's I-E-O model, Heaney and Fisher (2011) disclosed that *inputs* include information on personal characteristics and previous experiences. According to the authors, Astin (1993) referred to *environment* as all the experiences to which a student is exposed in college, focusing on academic and social integration. Furthermore, Heaney and Fisher (2011) declared that *outputs* include the students' behaviors, or outcomes, after exposure to the environmental factors. Pascarella, Pierson, Wolniak, and Terenzini (2004) cite the conceptual works of Astin (1993), Chickering (1969), Pascarella (1985), and Pascarella and Terenzini (1991), which suggested four types of influences considered to accurately estimate and understand the impact of college on students. According to Pascarella et al. (2004), the authors cited state that these influences are (1) student demographic or precollege characteristics, (2) organizational or structural characteristics of the institution attended, (3) students' academic experiences, and (4) students' nonacademic experiences.

Tinto's theory of student departure further attested to the validity of Astin's theory of student involvement. This model of student departure, also known as Tinto's theory of academic and social integration or Tinto's Integration Model, supported the

critical role of student involvement posited by Astin, and the impact that involvement has on student *persistence* (Milem & Berger, 1997). According to Bean and Eaton (2001), Tinto's 1975 model was built on Spady's adaptation of Durkheim's concepts, and this became the source for the academic integration (value congruence) and social integration (social support). Milem and Berger (1997) further highlight the fact that Tinto's interactionalist model of individual student departure, as it is also called, is quite similar to Astin's theory of involvement as they both deal with the issue of *persistence* in college. According to the authors, Tinto's 1993 revision of his 1975 conceptual model detailed the interaction between student behavior and perception as they integrate their social and academic environments. Hence, this framework guided this current study on student *persistence*.

Though other theories of student *persistence* exist, Cabrera et al. (1993) stated Tinto's Student Integration Model and Bean's Student Attrition Model as the two theories that have provided a comprehensive framework on college departure decisions. Having discussed Tinto's theory earlier, Bean's Student Attrition Model is another theory that was used to address student *persistence*. Cabrera et al. (1993) indicate that Bean's work builds upon process models of organizational turnover, and models of attitude-behavior interactions (behavioral intentions to stay or leave). According to Bean's theory, the intentions of students to persist are influenced by their attitudes and behaviors, which in turn affect their level of satisfaction with the institution (Jama, Mapesela, & Beylefeld, 2008). Bean and Eaton (2001) concur, citing attitude-behavior theory as providing the overall structure that leads to academic and social integration. Bean's model considered

the interactions of the student with the college environment as a sense of academic and social integration (Bean & Eaton, 2001). The authors emphasized that academic integration plays a vital role in the retention and graduation of students. Thus, student *persistence* comes into play.

Another important theory guiding this research is the Social Learning Theory (SLT) by Albert Bandura. In the research, Bandura (1971) grouped learning (1) by direct experience, based on the consequences of a given action, (2) through modeling, by environmental influences, and (3) by the reciprocal influences process, where there is interaction between behavior and its controlling conditions. In the SLT view, Bandura (1971) purports that individuals are neither driven by internal forces nor controlled by environmental influences, rather by the interaction between their behavior and its controlling conditions. Nabavi (2012) stated that SLT has become the most influential theory of learning and development. The author affirmed that it is rooted in many of the basic concepts of traditional learning theory. According to Nabavi (2012) and McLeod (2016), the theory has often been called a bridge between behaviorist learning theories and cognitive learning theories because it encompasses attention, memory, and motivation. Culatta (2018) alluded to this, indicating that SLT explains human behavior in terms of continuous reciprocal interaction between cognitive, behavioral, and environmental influences. Additionally, Mcleod (2016) stated that due to its cognitive focus, Bandura modified his Social Learning Theory (SLT) to Social Cognitive Theory (SCT), as a better description of how individuals learn from their social experiences.

Nabavi (2012) highlighted that this theory expansion provided a framework for understanding, predicting, and changing human behavior.

The Theory of Educational Productivity by Walberg (1981) was used to highlight the institutional factors that were employed in this study, thus addressing *academic success*. Under the theory, domains of variables were identified which were most influential to educational productivity. In Walberg's theory, nine factors require optimization to increase affective, behavioral, and cognitive learning (Walberg, 1984). According to Walberg (1984), these factors fall into three groups - aptitude, instruction, and environment. The first set, aptitude, included ability and prior achievement as measured by standard tests, their development indexed by age, and personal motivation as indicated by students' perseverance or *persistence* to learn. Instruction highlighted the amount of time a student spent on learning. Environment included climates of the home, classroom groups and peer groups outside school, as well as leisure time. Additionally, McGrew (2007) took note of authors who demonstrated the importance of the domains of motivational orientations, self-regulated learning strategies, and social/interpersonal abilities in facilitating academic performance or *academic achievement.*

Another evolving framework considered in this study is the learning theories conceptualized by Edgar (2012). Highlighting learning theories in education, Edgar (2012) emphasized *behaviorism* as the basic tenet underlining the theories, building on John Watson, B.F. Skinner, and Ivan Pavlov's behavioral theories. Similar to Walberg's theory, Edgar (2012) conceptualized a learning environment with four design perspectives, namely, Learner-centered (skills, attitudes, and beliefs learners bring to the

educational setting), Knowledge-centered (enabling students to understand and transfer knowledge), Assessment-centered (providing opportunities for feedback and revision congruent with learning goals), and Community-centered environments (encouraging the degree to which learning environments promote a sense of community). Edgar (2012) cited aspects of community-centered environments as the classroom, the school, and the connection between the school and the larger community, as well as the home. Furthermore, the understanding of community importance and its effect on learning can have a significant impact on academic achievement (Edgar, 2012).

These emphasized theoretical and conceptual frameworks serve as the groundwork for the current study and provide an understanding of the various factors that influence students' p*ersistence* and success in their academic pursuits. They further serve to highlight the selected demographic and institutional factors impacting students, and how these factors ultimately predict their *persistence* and *academic success* as international students.

Research Hypotheses

The predictive power of selected demographic and institutional factors on the p*ersistence* and *academic success* of the international students was discussed. Thus, the following hypotheses were generated for this study:

H_1: There is a statistically significant relationship between the demographic factors (gender and ethnicity) and persistence among international students.

H_2: There is a statistically significant relationship between the institutional factors (campus interactions, school climate, support services, and social activities) and persistence among international students.

H_3: There is a statistically significant relationship between the demographic factors (gender and ethnicity) and academic success among international students.

H_4: There is a statistically significant relationship between the institutional factors (campus interactions, school climate, support services, and social activities) and academic success among international students.

Assumptions

Several assumptions were made in this study. These are listed below:

1) The researcher assumed that responses to survey questions would be as accurate as possible, since study participants were to provide data anonymously.
2) With anonymous participation, it was assumed that demographic and academic data would be reported accurately by the study participants.
3) It was assumed that international students under 18 years of age would not respond to the survey. Participants recruited for the study were international students who were 18 years and older.
4) The researcher assumed there would be diversity in ethnicity as the study participants come from all over the world.

5) It was assumed that the sample study participants would be a representative of the population.

Limitations/Delimitations

Due to the broad nature of the study, there was the need to place the research under specific confines. To make the variables in the study measurable, and the findings generalizable, boundaries and parameters were set around the study. The decision to undertake this study was based on researcher interest and empathy with the selected study population. This interest prompted the selection of the specific demographic and institutional factors to determine their predictive power on the *persistence* and *academic success* of the international students.

Even with the set boundaries, there were certain limitations.

1) Primary demographic and academic data, as well as current GPAs for the participants, were not assessable; so, self-reported data from the survey instrument had to suffice.
2) Study participants were not mandated to respond to all statements, limiting the number of complete and usable data. This ultimately impacted study results.
3) Research approval for this study came from the Institutional Review Board (IRB) at a time when most potential study participants were not able to conveniently be part of the study.

4) Low turnout on the initial data collection days rendered the collected data not immediately usable. This brought about more than one data collection day.

5) Findings could only be generalizable to the population of international students in the chosen institution.

Nonetheless, the research could be extended to other educational institutions, or institutions of higher learning, with similar populations.

Definition of Variables/Terms

In this study, various terms were used to highlight the design and concept. For the purpose of this paper, the terms are indicated and explained below:

Academic Success: This is the outcome variable studied. It was measured on a 4-point scale by participants' GPA at the time of data collection. The term is interchangeable with *student success* and *academic achievement*.

Campus Interactions: It is the frequency of interaction between the participants and selected individuals on the school campus on a weekly basis. This is measured on a 5-point scale from "never" to "always."

Demographic Factors: An independent variable in the study. Categorized as *gender* and *ethnicity* of the international students who were the study participants.

Ethnicity: The four major ethnic groups (African Descent, Asian, Middle Eastern, and "Other") that the international students fell into in this study. It is one of the categories under demographic factors.

Gender: A category under demographic factors. The international students are either male or female.

Institutional Factors: One of the independent variables in the study. The institutional factors were categorized as *campus interactions, school climate, support services,* and *social activities*.

International Student: Student who left his or her home country and is currently enrolled in the selected institution where the study took place. The person is identified as a participant in the study. This term is interchangeable with *foreign student*.

Persistence: It is another outcome variable studied. It is measured by the participant's indication to return to the same educational institution the following semester.

School Climate: The overall perception of the campus environment by respondents. This is measured on a 5-point scale from "strongly disagree" to "strongly agree."

Social Activities: Campus activities available to international students. This factor is measured on a 5-point scale, in "hours per week," from "less than 1" to "more than 15."

Support Services: Selected services perceived by participants as promoted in the institution where they are enrolled. It is also measured on a 5-point scale from "never" to "always."

Organization of the Study

The layout of this research study was mapped into five chapters. Chapter 1 consists of the Introduction, the Statement of the Problem, the Significance of the Study, the Theoretical Framework, the Research Hypotheses, Assumptions, Limitations and Delimitations, as well as the Definition of Variables and Terms. Chapter 2 outlined the Review of the Related Literature regarding student *persistence* and *academic success.* Chapter 3 focused on the Methodology used in the study. This included the Type of Study Design, the Population and Research Setting, Sampling Procedure, Instrumentation, Validity and Reliability of the Instrument, Data Collection Procedures for Pilot Test and Field Study, Identification of Independent and Dependent Variables, the Null Hypothesis, Statistical Analysis, and Evaluation of Statistical Assumptions. Chapter 4 exhibits the Analysis and Evaluation of Collected Data, including the explanation of results. Finally, Chapter 5 offers a summary of Findings in the Study, Discussions, Study Implications, Recommendations for Future Research, and the Conclusions.

CHAPTER 2

LITERATURE REVIEW

Overview

The main question posed in this study was whether selected demographic and institutional factors predicted student *persistence* and *academic success*, especially for international students. Recall that Fischer (2014) reported research across the nation where there were concerns in various educational settings regarding the increased rate at which international students leave institutions without graduating. This has led researchers to initiate ways of verifying this trend and the impact it may have in different sectors of the education industry. Research could identify practices that can lead to the greater effectiveness and efficiency of international student recruitment and retention practices. Dancer, Morrison, and Tarr (2015) view effectiveness as improved overall grades, grade distribution (less fails, more credits/distinctions), or improvements in retention and progression in degree plans. Improving *persistence* and *academic success* could also lead to greater effectiveness in the education industry.

Several studies have examined students' *persistence* and their *academic success*. Each study viewed the phenomena based on varying factors and conditions, while investigating them using different methodologies. As indicated earlier, there were two independent variables (*demographic* and *institutional* factors) in this study. Furthermore, the demographic factors were categorized as *gender* and *ethnicity*, while institutional factors were categorized as *campus interactions, school climate, support services,* and

social activities. The dependent variables were student *persistence* and their *academic success*. The review of related literature was discussed in this chapter, with reflection on studies that investigated student *persistence* and their *academic success.* These studies were not pursued though, in the light of the variables in this present study.

International Student Persistence and Academic Success

With the issue of diversity stemming from the influx of international students into the U.S. education system, many universities are facing the challenge of retaining this population and helping them to attain *academic success.* According to Perry et al. (2017), studies that discuss potential issues with current international students provide important demographic data, college choice, prevalent challenges, and elements of student satisfaction. Thus, Perry et al. (2017) recommended studying the opinions and perceptions of international students in order to develop and maintain programs leading to higher enrollment and retention rates for this population of students. To attest to this, Mamiseishvili (2012) carried out Pearson's chi square tests to examine significant association between study variables and international students' *persistence*. Furthermore, logistic regression was implemented to determine the specific factors that influenced their *persistence*.

On the issue of *academic success,* several studies investigated the phenomenon as it related to international students. Perry et al. (2017) noted that international students attain *academic success* at a higher level than their American peers, even as they bring significant financial growth to the American economy. In the various studies, *academic*

success was defined operationally as semester GPA, cumulative GPA, admissions scores, and scores on tests and coursework. Notably, York et al. (2015) did a comprehensive literature search to help define and measure *academic success*, and hierarchical regression analyses were conducted to analyze the main study hypothesis with GPA as the dependent variable. No matter the way the phenomenon was addressed, questions regarding the *academic success* of international students would arise. Thus, investigating selected factors that impact students' *academic success* as international students was important, and reviewing the related literature was advisable.

Demographic Factors, Persistence and Academic Success

Demographic factors in this current study are *gender* and *ethnicity*. With the cross-cultural differences, and domestic roles played out in the various cultures, it was imperative to determine how *persistence* and *academic success* would be predicted by *gender* and *ethnicity*. In a longitudinal study, Mamiseishvili (2012) utilized the Beginning Postsecondary Students (BPS:04/06) data set to examine the national profile of first-year international students in U.S. postsecondary institutions. With 23,090 students from 1,360 postsecondary institutions observed, the author centered the study variables on students' demographic attributes and in-college characteristics, as well as academic and social integration measures. However, during the investigation, Mamiseishvili (2012) noted limited data on the *persistence* of international students.

Gender

Using descriptive statistics, Mamiseishvili (2012) identified the participant demographic characteristics. Mamiseishvili (2012) found that the average age in the study was about 21 years, and there were more men (54.7%) than women (45.3%). Young and Snead (2017) investigated gender-related issues in the academic society, and differences in the Saudi Arabian and American culture as they allow men and women to travel outside their country to study. The action research process was used in the qualitative study to examine and analyze the participants' responses. Utilizing multiple forms of data for triangulation, coding, and analysis, Young and Snead (2017) identified emerging themes and patterns of thought. With consent, a segment of the participant population was surveyed, participated in a focus group discussion, wrote personal reflections, and was interviewed (Young & Snead, 2017). Also, the researchers noted that all participants completed a short written demographic survey in English, regarding gender, major, and country of origin.

The overall findings in the research by Young and Snead (2017) revealed that the participants, regardless of gender, experienced challenges and barriers to success during their acculturation process. According to Young and Snead (2017), all participants reported cultural and social differences that were gender-related, involved the friendliness of Americans, and related to the availability of student activities and academic programs, social, language, and academic progress difficulty, as well as a lack of support services for non-English speaking students. In another study, Curtin, Stewart, and Ostrove (2013) examined potential differences between international and domestic doctoral student

experiences in graduate school. Findings by Curtin et al. (2013) indicated that the international students were more likely than their domestic counterparts to study in the physical sciences and engineering, were more likely to be male and of Asian background, and finally were less likely to have financial struggles.

With regards to their financial aid status, Sanchez and Smith (2017) investigated the predictive power of gender, GPA, credits earned, credit status, and financial aid amount, on the *persistence* and graduation status of international students. Sanchez and Smith (2017) conducted a quantitative, exploratory analysis of 535 community college students in a Hispanic Serving Institution (HSI) to determine their eligibility or ineligibility for Federal Student Aid (FSA) as non-U.S. citizens. According to the authors, data analyzed were obtained through the Free Application for Federal Student Aid (FAFSA), where applicants were asked to confirm their citizenship status.

In their study, Sanchez and Smith (2017) found that 45% of the students were male and 55% were female. The authors noted that 40% of the students were eligible for FSA, while 60% were ineligible. Sanchez and Smith (2017) further indicated that 80% of the students persisted, while 20% did not. Findings by Sanchez and Smith (2017) also indicated that gender differences were not present. However, the authors noted the non-significance of Federal Aid status by GPA, meaning that students performed equally regardless of their FSA status (Sanchez & Smith, 2017). Perry et al. (2017) posited that age and gender affected information use among international students, recounting an e-mail survey where higher-level students needed information such as databases, remote access to library offerings, and e-journals to support their academic courses.

Navarro, Flores, Lee, and Gonzalez (2014) conducted a longitudinal study on the *persistence* of engineering students in an HBCU using the test-retest approach. The authors used the Social Cognitive Career Theory (SCCT) developed by Lent, Miller, Smith, Watford, Lim, Hui, et al. (2013) as their framework. According to Lent, et al. (2013), the SCCT model was created to determine the development of careers and goals that attract students with regard to engineering. Thus, to assess students' academic satisfaction and *persistence*, Navarro et al. (2014) tested the unidirectional, bidirectional, and unidirectional integrative SCCT models along with the reciprocal paths.

Study results by Navarro et al. (2014) showed that gender was a model fit for the data in the bidirectional model. However, the authors reported no substantial differences in the models, thus claiming that gender did not significantly moderate *persistence*. The study by Stewart, Lim, and Kim (2015) used an ex post facto design to test the hypotheses about the impact of selected factors on *persistence*. Examining group differences in *persistence* using factorial ANOVA, the study compared students placed or not placed in remedial courses, as well as any interaction effects. The study findings revealed that gender had no statistically significant main effect on *persistence* ($p = .528$), and that, overall, remedial status of students did not depend on gender ($p = .302$).

Ethnicity

As individuals thirsting for knowledge, international students have things in common with their American counterparts. Nonetheless, ethnicity and cultural differences may impact their academic journey, especially with no close support system

available. According to Perry et al. (2017), students must deal with the differences in American culture which they are often not familiar with. Perry et al. (2017) studied 862 international students at a Western university in the United States and collected demographic and group comparison information. The investigators ran Cronbach's alpha reliability analysis to ensure that the survey's items were a reliable indicator of intended measures. With a repeated t-test, which led to three paired t-tests, two paired t-tests, and an independent samples t-test, the investigators found that the greatest concerns of international students were financial, health, and language issues. Perry et al. (2017) noted that the language barrier can be a complicated issue for many students. The researchers emphasized that this barrier has been particularly troublesome for Asian students whose languages lack a direct English translation.

Focusing on ethnicity, Mamiseishvili (2012) noted that 50.3% of students identified themselves as Asian, 20.1% as white, and 13.1% as black. The author further noted that 12.1% were Hispanic or Latino, and 4.4% were identified as other race or checked more than one race. In terms of race/ethnicity, Mamiseishvili (2012) found a positive and significant relationship between language proficiency and academic performance, supported by significant effects of GPA and remedial English on *persistence*. Noting the difference between international students who required remedial English in their first year and those who did not, Mamiseishvili (2012) suggested that international students who come to the U.S. with better academic preparedness and language skills are more likely to persist than students who lack language proficiency and academic preparedness.

Regarding discrimination, Perry et al. (2017) found that international students deal with the issue by preferably cultivating friendships with those from nations and cultures similar to their own. Findings in their study agreed with previous studies where language had been previously linked with discrimination. In the interviews conducted as part of the research by Sullivan (2018), the international students responded that language was one of the major issues that they had to deal with when adjusting to campus situations. Sullivan (2018) stated that the students reported experiencing initial difficulty in classes, social situations, and when seeking resources on campus. On students expressing differences between their speaking and writing the English language, Sullivan (2018) further acknowledged that the complexity and severity of the situation depended on the international student's level of English language proficiency and the context of the conversation. Authors noted that more research can be conducted on this issue. In the study by Sullivan (2018) regarding the placement of international students in the same room with American students, the researcher noted that students who were given this type of room assignment had experiences related to language, culture, and cleanliness. Sullivan (2018) further noted that the international students overall, found that the experiences assisted them in adjusting to campus life and English language proficiency.

On another note, Navarro et al. (2014) also used the SCCT academic and career model developed by Lent et al. (2013) with regard to *persistence* in selected ethnic groups. Here, model-to-data fitness was also determined. Results in the study showed that race/ethnicity fit perfectly in the bidirectional model. However, the variable did not significantly moderate *persistence*.

Institutional Factors, Persistence and Academic Success

Institutional factors expressed in this current study were *campus interactions, school climate, support services,* and *social activities*. As part of campus experiences, international students interact daily with different individuals and personnel, and have a sense of the school environment. Furthermore, they have access to support services and social activities for academic/wellness balance. Related literature regarding these factors was hereby discussed.

Campus Interactions

Amirali and Bakken (2015) investigated the trends and challenges of recruiting and retaining international graduate students in a U.S. university. For feedback, the authors surveyed graduate students, collecting data from prior and current international graduate students. They also evaluated the past trends and activities implemented in the institution and analyzed current office procedures. Their results indicated no real communication with students once they were notified that they were admitted. Other results show international students reporting no communication prior to arrival on campus, and no help in getting to campus to begin their education. Sullivan (2018) further noted the language challenges that students had, especially those with thick accents. Thus, the challenges to the international students centered on the negative interactions with individuals and units on campus.

Amirali and Bakken (2015) reported in their research that the graduate school developed the 3Cs strategy (Connect, Communicate, and Collaborate) to benefit

international graduate students. According to the investigators, the metrics used to determine the project effectiveness and success in campus interactions were percentage growth in number of referrals from current students and alumni, as well as the results from the Quality of Life survey. So far, the authors noted, the institution's commitment to connect, communicate, and to interact with the international students has brought positive results. Findings by Mamiseishvili (2012) highlighted the importance of study groups and peer interactions outside the classroom, and further noted that validation from faculty, staff, and peers, was critical for international students and their adjustment in college. Sullivan (2018) cited the peer-to-peer interaction within international clubs, where international students had immediate opportunities for involvement on campus, making their transition easier by spending time with students who shared common experiences.

Perry et al. (2017) acknowledged that international students who collaborated and made friends with American students were able to adjust and adapt better than those who did not. On the other hand, Curtin et al. (2013) found that international students did not report positively regarding greater advisor support, though, the researchers noted, this was an area where the international students may have greater needs than their American counterparts. According to Perry et al. (2017), students can benefit from the skills of their advisors and faculty who understand the academic changes in the students. In their study regarding international students in engineering programs, Zhang and Dinh (2017) noted challenges encountered by these students and emphasized the significant role that academic advisors played in their *academic success*, especially with constant and continued interaction. Zhang and Dinh (2017) focused on academic advising improving

the learning experiences for international engineering students, though this could apply to all international students. Newell (2015) concurs, narrating the role of advising as important also to international student athletes. Zhang and Dinh (2017) further noted other researchers who have indicated that academic advising played a pivotal role in student retention, personal development, and *academic success*.

Participants in the study by Sullivan (2018) indicated positive responses regarding their interactions with their professors. Sullivan (2018) stated that students had positive responses regarding helpful professors who repeated terms and further explained confusing concepts related to language differences. According to Sullivan (2018), eight of the students gave specific examples of professors taking time to attend to them in and out of the classroom, answering questions, clarifying lectures, and showing interest in their *academic success*. Interactions with professors, Sullivan (2018) continued, went as far as attending athletic competitions, and visits by the students to their homes. Zhang and Dinh (2017) added research citing that international students were more likely than domestic students to view a close relationship with advisors as important toward their *academic success*. Zhang and Dinh (2017) further cited studies where findings demonstrated that satisfaction with academic advising significantly predicted students' sense of belonging and intent to persist.

School Climate

The transition to a new environment always poses a lot of challenges to students, more so to international students. The school climate experience for the international

student can include the classroom environment, people they meet daily on campus, access to tutoring, advisors, and socialization with different ethnic groups. According to Amirali and Bakken (2015), international graduate students come from diverse backgrounds and the transition to a new environment is challenging, especially as it pertains to culture, communication, accommodations, and the U.S. education system, to name a few. With respect to recruitment and retention of international graduate students, the researchers found that while universities are challenged with adopting better recruitment strategies in application verification, prospective international students face more complex hurdles, as they are unlikely to visit a campus before deciding whether to attend.

In a U.S. university they investigated, Amirali and Bakken (2015) further stated that international students stressed a not-so-friendly environment. The researchers noted that students reported that no help was given to them on campus once they arrived, nor when they were seeking accommodations. With these first impressions of a campus environment, international students may not stand the chance of persisting in that institution or achieving *academic success*. Using the 3Cs strategy for their project, Amirali and Bakken (2015) stated that *student success* was based on the overall performance and increase in most of the metrics as indicated earlier. Attesting to the climate in classrooms, Perry et al. (2017) stated that international students can contribute to classroom discussions, projects, and other activities, creating much needed diversity in opinions usually lacking there. Mamiseishvili (2012) concurs, indicating that GPA, degree goals, and academic integration had significant positive effects on *persistence* of undergraduate international students.

Regarding access to tutoring, Sukhlecha, Jadav, Gosai, and Balusamy (2016) stated that, at present, due to faculty shortage and lack of active and complete participation by students, it is becoming very difficult to conduct conventional tutorials (CT). In their study, they hypothesized that in tutorial groups, there is a reinforcement of prior knowledge, information recall, reasoning, theory building, conceptual change, and collaborative learning. Sukhlecha et al. (2016) presented pharmacology lectures, followed by a pre-intervention test for a batch of the 2nd year medical undergraduates. As the researchers reported, the students were placed in either the Student-Led Objective Tutorial (SLOT) as the interventional group A or the CT group B as control. The investigators reckoned that students in the SLOT group prepared five multiple choice questions on the given topic and presented these to other teams and audiences. Sukhlecha et al. (2016) further recalled that proceedings were facilitated by two lecturers in the CT control group, after which a post-intervention test was given to both groups, and feedback sought from students and teachers on Student-Led Objective Tutorial (SLOT).

The Findings by Sukhlecha et al. (2016) showed that the mean marks in Group A improved by 31%, and in Group B they improved by 11%. According to the authors, academic performance following the SLOT intervention was better than the conventional tutorial (CT) control. Sukhlecha et al. (2016) noted that 63% of students favored SLOT, as it not only stimulated their interest in the topic, but also improved their self-learning skills and teamwork. The authors established that the teachers also favored SLOT for similar reasons and concluded that it was an effective alternative to CT and promoted active learning among students through group work, leading to greater satisfaction and

better performance on tests. They also stated that SLOT helped where there was faculty shortage, with student learning influenced by the quality of discussion, ultimately bringing success in studies.

Reviewing further the school environment and its impact on *student success*, Perry et al. (2017) noted in their study that many international students identified academic issues as more important than friends and social activities. In agreement, Curtin et al. (2013) purport that, compared to domestic students, international students placed a higher value on research-related and other academic experiences, and reported a stronger sense of belonging. Noting these academic environments, Perry et al. (2017) thus stated that suggestions have been made urging universities to form study groups, which would assist international students in their coursework.

Support Services

Support services, an important issue in the academic progress of students, must be acknowledged. One can see that study time, extra learning services like tutoring, and mentoring services are important. Other support services available to the international student could be in the form of wellness services, non-academic help, and advisor availability. Perez-Encinas and Ammigan (2016) investigated international students and their level of satisfaction with support services as a means to ensure a positive student experience. The authors compared the international student satisfaction tools used in Spain and the United States (U.S.), thus comparing the Spanish and English-speaking countries. Five of the most-widely used tools presented by the authors were (1) The

International Student Barometer (ISB), (2) Ruffalo Noel Levitz, (3) StudyPortals, (4) National Survey of Student Engagement (NSSE), and (5) QS Student Satisfaction.

Furthermore, Perez-Encinas and Ammigan (2016) conducted a pilot study to assess and analyze issues related to support services in the institutions. The researchers invited 40 institutions from Spain and the U.S. to participate in a 15-item, anonymous online survey developed around demographics, type of support services, effectiveness, and usefulness. Additionally, they carefully selected survey items to provide an overview and assess the main types of support services used in both countries. Fifteen institutions from Spain and 12 from the U.S. responded to the invitation. Reviewing the role of support services, Mamiseishvili (2012) recognized the role of academic advising in successful retention of international students.

In their study, Perez-Encinas and Ammigan (2016) found that support services provided by institutions in each country were defined by and aligned with the type of student mobility, whether credit as in Spain or degree as in the U.S. Furthermore, in the U.S., international student support offices focused on providing programs that promote *academic success*, acculturation sessions, as well as campus and community engagement, among other services. According to the authors, feedback enabled university officials to implement necessary changes in the areas of customer service, student advising, programming, outreach, and educational training. Thus, discussions from their study signified the positive influence of support services on *academic achievement*.

Sanchez and Smith (2017) studied non-U.S. citizen student differences in Federal Aid, and demonstrated that eligibility for financial aid could help students get admission

into an institution, carry higher credit loads, and persist to the next year. The authors declared that it was not so for non-U.S. citizen students who are ineligible for federal student aid (FSA). The research by Sanchez and Smith (2017) highlights the need for such support structure, to reduce the gap toward *persistence* within non-U.S. citizen students. Furthermore, the authors indicated GPA as a significant predictor of *persistence*, noting the importance of continued student services programs augmentation to ensure that support systems were available for increased academic achievement by all students. For policy and decision-makers who create and revise award programs, Sanchez and Smith (2017) remarked that the financial aid amount awarded should be considered as it also contributed to *persistence.*

Despite the similarity in achievement, another unique finding by Sanchez and Smith (2017) was that students who were ineligible for FSA received less funding for their education at statistically significant levels. Furthermore, the authors declared that these students also enrolled in, and earned fewer credits, and did not have similar levels of *persistence* with those who were eligible for FSA. Regardless of FSA status, Sanchez and Smith (2017) noted, students without funding did not persist at similar levels to their counterparts with funding. This highlighted the importance of financial aid for students, particularly for a vulnerable population, such as those who are ineligible for FSA due to their non-citizen status.

In other research targeting support services, Singaram, Van Der Vleuten, Muijtjens, and Dolmans (2012) investigated the perceived overall productivity of tutorial groups and student academic achievement using structural equation modeling (SEM).

Singaram et al. (2012) hypothesized that English as the First Language (EFL) and higher secondary school scores will have a positive impact on student academic achievement. The authors conducted a path analysis to test a causal model in which EFL and secondary school scores were assumed to influence process variables, and these variables were assumed to influence overall group productivity and, ultimately, students' academic achievement. In this study, students worked in tutorial groups of 10-12 students, around clinical and basic science content themes in six to eight-week rotations, with a limited number of lectures, practical sessions and skills trainings. Furthermore, together with a facilitator, 387 medical students met twice a week for a two-hour session to discuss problem-based learning (PBL) paper cases, based on the Tutorial Group Effectiveness Questionnaire (TGEQ).

In the findings by Singaram et al. (2012), the causal model demonstrated that EFL and the secondary school score positively affected the student academic achievement score. The researchers stated that demotivational group processes negatively affected overall group productivity (beta = -0.25) as well as the academic achievement score (beta = -0.15). Singaram et al. (2012) concluded that the proportions explained variance (R-square) of 0.15 for academic achievement and 0.45 for overall group productivity, indicated that English as the First Language (EFL) and secondary school scores had a positive effect on academic achievement.

With a focus on international students, Dancer et al. (2015) employed micro-level data to investigate the effects of students' attendance in peer-assisted study sessions (PASS) program on their final grade in their first-year business statistics course. Ordinary

least squares, random effects, and quantile regression models were used to model the data. According to Dancer et al. (2015), results demonstrated that PASS participants clearly had a significantly higher average mark in business statistics compared to non-PASS participants. The investigators also reported significant findings for institutions that were implementing similar programs. Furthermore, Dancer et al. (2015) noted related findings where institutional efforts to strengthen and improve student retention were accessible, thus, effectively supporting international students.

To further attest to support services, Sullivan (2018) highlighted academic resources noted by the international students as the library which was part of their campus experience and critical to their academic success. Tutoring was also a recurring theme amongst them, the researcher reported. Regarding the International Students Office as a support service, Sullivan (2018) found that this unit provided an array of resources that helped welcome the international students to the campus. According to Sullivan (2018), resources ranged from basic toiletry kits and linens to winter clothes. These were important resources, as the students did not have funds to adequately begin their semester away from home.

On the other side of the spectrum, Sullivan (2018) identified a theme that was a critical component of the campus experience for the students. According to Sullivan (2018), the colleges researched were in rural areas and not well equipped to provide public transportation, close retail shopping experiences, restaurant varieties, or safe sidewalks. Sullivan (2018) stated that these issues reinforced the importance of campus resources for the international students, noting their frustration as they discussed their

inability to navigate the community, take care of personal business, like doctor appointments, or simply go shopping for needed personal items.

Social Activities

In the study by Sullivan (2018), the students identified campus-provided activities as playing an important part in their campus experiences. For these students, Sullivan (2018) noted, the activities planned by the International Students Office and those provided by the campus activity boards impacted their experiences. Sullivan (2018) stated that the activity boards were the primary resources for international students to enjoy outings, the beach, camping, music, dance, movies, cultural events, socializing, and leadership opportunities. This, the researcher stated, was a means of introducing international students to their American counterparts, as well as helping them learn more about the American and campus culture.

Sullivan (2018) recounted how the colleges researched had flexible housing plans for students during breaks, where some students returned to their home country, others went home with American friends, while some stayed on campus. According to Sullivan (2018), first year international students expressed their frustration about not having access to apartments with kitchens. Sullivan (2018) exclaimed that such access provided these students additional opportunity to meet other international students on campus and offered them social experiences over cooked meals. In the study by Mamiseishvili (2012) on the other hand, social integration measured by school clubs, sports, and fine arts activities had a negative effect on *persistence*. The findings, according to Mamiseishvili

(2012), could be that the international students were involved in other kinds of social activities such as multicultural and international events not captured in the study.

Among the campus activities discussed in the study by Sullivan (2018), the program developed and sponsored by the International Student Offices was identified as an important part of the campus experience for the students. Also, Sullivan (2018) stated that students identified the international student clubs as a major student activity programmer on the campus. According to Sullivan (2018), the colleges reported that the international office programs supplemented the larger student activity programs on campus. In one of the colleges researched by Sullivan (2018), international students produced a major show each year, showcasing international talent to the entire campus and local community. From the students' perspective, Sullivan (2018) hinted, such international programs resources were important, as they often provided the students with unique experiences and helped with their transitional and cultural adjustments on each campus. Meanwhile, Mamiseishvili (2012) suggested the need for international admissions offices, funding agencies, faculty, academic advisors, English language program staff, and student affairs professionals to work together in an effort to identify and recruit international students who are determined to obtain U.S. degrees. This collaboration promoted *persistence*.

Summary

As an important segment of the student population, researchers have begun to investigate international students and their activities in educational institutions. In their

studies, the researchers identified factors that potentially were challenging to students' *persistence* and *academic success*. Studies and literature are now identifying language barriers as one of the most significant challenges among international students, as well as the need for support services. Other studies reviewed investigated the school climate, international students' perspective on the campus environment, their interactions with various individuals, and the social activities provided for entertainment on the campuses. However, the studies do not collectively identify the demographic and institutional factors highlighted in this study as impacting the *persistence* and *academic success* of these students.

With recruitment and retention goals as the starting point, other additional predictions of student academic performance may be identified. Early monitoring of these predictors was one good way for educators to detect achievement difficulties and plan appropriate remediation. Nonetheless, the factors investigated did not include any challenges in terms of the breakdown of demographic and institutional factors in this present study. Looking into the related literature above, it was thus necessary to address the selected factors and identify relationships, with the goal of examining their predictive power on the *persistence* and *academic success* of international students. Thus, there was the need for this current study.

CHAPTER 3

METHODOLOGY

The review of related literature showed different ways that student *persistence* and *academic success* of international students were measured and several ways that the researches were carried out. This present study identified selected demographic and institutional factors and investigated how they predicted the *persistence* and *academic success* of international students enrolled in one of the Historically Black Colleges and Universities (HBCUs) in an urban metropolitan area in the southern part of the United States. None of the previous studies determined the *persistence* and *academic success* of international students based on the predictive power of these selected demographic and institutional factors. That is the objective of this study. The methodology of this current study is discussed in this chapter. The discussion regarding methodology, inter alia, addressed the research design, instrumentation, validity, and reliability, as well as the identification of variables and statistical analyses. In the preliminary stage, a pilot study was carried out to test the hypotheses and determine if the study could go on.

The Research Design

Two research designs were employed for this study. These were the *Survey* and the *Correlational* research designs. The Survey research design involved using the cross-sectional survey method. According to Gay, Mills, and Airasian (2012), a cross-sectional survey is one in which the researcher collects data from selected individuals once at a

single point in time. In this study, data regarding participants were generated and collected using the *Persistence* and *Academic Success Inventory* (PASI) questionnaire. This survey tool was used to collect and identify participant responses and group items for analysis. The PASI questionnaire was designed as an original measuring instrument having three sections highlighting various items. Section one comprised items pertaining to the general demographics of the participants. The second section had items assessing the institutional factors and their predictive power on the *academic success* of the participants. In section three, the items assessed the *persistence* of the participants. This section also addressed the overall expectations by the participants, with an avenue for them to indicate comments, feedback, and biases encountered. Amendments to the instrument were made after reliability and validity were confirmed through pilot study and field test.

The other research design for this study was the Correlational research. According to Gay et al. (2012), the Correlational research design involved the researcher collecting data and attempting to identify if there was a relationship between two or more variables in the study, and to what degree. Gay et al. (2012) stated that a Correlational study may be used to determine relations among variables (relationship study), or to use these relations in making predictions (prediction studies). Reasons behind the relationship between these variables are thus ascertained through these predictions. Gay et al. (2012) further noted that the degree of relationship is expressed as a correlation coefficient. According to Gay et al. (2012), Correlational research studies typically involve obtaining

scores for two or more variables of interest from each participant in the study, pairing the scores, and then correlating them to one dependent variable.

In this study, participants were assessed on demographic and institutional factors to determine their predictive power on each of the outcome variables of *persistence* and *academic success.* The variation of the Correlational research approach used was the Prediction Study. The ultimate goal of the study was to see whether the selected demographic and institutional factors would influence the decision of international students to return to the school the following semester (*persistence*) or their GPA (*academic success*). With this process, the participants were available to respond to the questionnaire, and the researcher did not need to wait for the effect to occur because every response was current at that point in time (i.e. decision to leave the school and semester GPAs were self-reported and available).

The Correlational research design was appropriate because the participants under study already fell into demographic categories given by their responses. For the *institutional* factors, participants' perceptions of these factors enabled them to give the responses that they gave, which drove the rating scores that were assigned to the factors. Scores obtained from each participant were correlated on the bases of the demographic and institutional factors. Though no possibility of randomization or manipulation existed as in the case of a true experiment, correlation of scores was still possible. See Figure 1 for the layout of the study design.

Research Design Layout

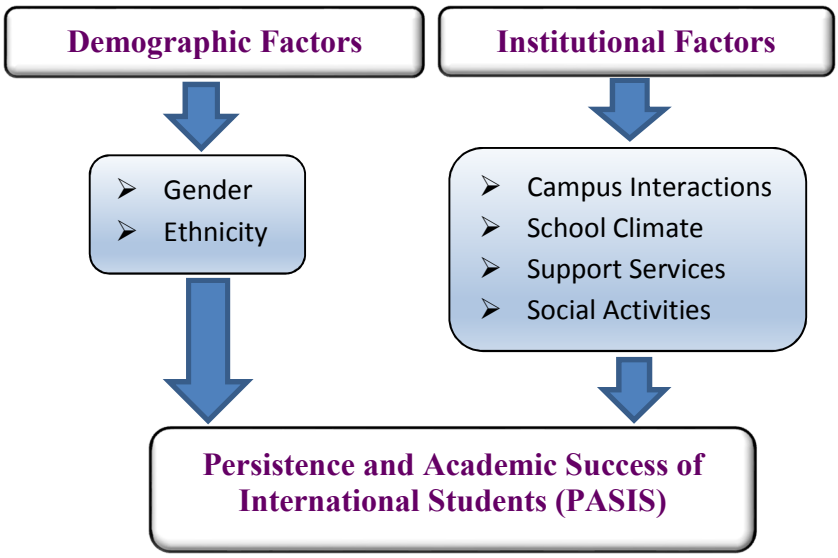


Figure 1. Predictive Power of Demographic and Institutional Factors on the Persistence and Academic Success of International Students

Population and Research Setting

The population surveyed in this study was the international students enrolled in one of the Historically Black Colleges and Universities (HBCUs). These students, by their default status in the United States, became potential study participants. Invitation for the study was made through the school database maintained by the International Students Office. Participants were also invited through international students' organizations, events, workshops, and class programs. These international students consisted of male and female students who have come from other countries such as Africa, Asia, Europe,

and the Middle East to further their education in the U.S. With the enrollment of these students, survey data illustrated the possible diverse nature of the population. For the participants in the study, there was no unique identifier or collector such as a student ID number. The students remained totally anonymous, with no follow-up information requested. The demographic breakdown of the population is noted in the report for the study.

The research setting for the study was the campus of the four-year HBCU. The campus is large and sits in the heart of an urban metropolitan area in the southern part of the country. The campus of the four-year college has several school buildings which are necessary infrastructure. However, online and in-person participation were available for the students on the college campus. Classrooms were used as actual venues during international workshops or events. The research setting in this current study was made close in nature to the everyday environment that the participants were comfortable with. For the study, effort was made to carry out face-to-face survey in a uniform research setting. The students who chose to participate from outside the school campus were informed of the research expectations.

Sampling Procedure

This current research investigated selected demographic and institutional factors, and their predictive power on the *persistence* and *academic success* of international students as identified in the study. Due to the nature of the study, the sampling procedure was in two stages. First the whole population was selected, because at any given time, a

finite number of international students are enrolled. These students were then classified under freshmen, sophomore, junior, and senior for the undergraduate programs, while those in the graduate program were classified under masters and doctoral. Non-probability sampling, specifically *Purposive Sampling*, was then used in the next step. According to Gay et al. (2012), purposive sampling, also referred to as *judgment sampling,* is the process of selecting a sample that is believed to be a representative of a given population. Only one unique group was invited to participate in this study, and any student who took part in the study represented the population.

Furthermore, it became obvious that some international students enrolled were exempt (participants must be 18 years or older to be in the study). After the sampling process, the survey request was given to all eligible international students. Since it was an anonymous survey and voluntary, not all international students took part in the study.

Instrumentation

Cognitive, Affective, and Projective tests are the most commonly used measuring instruments in educational research (Gay et al., 2012). For this study, the measuring instrument was the affective test. An affective test is an assessment designed to measure mental characteristics related to one's emotions, such as attitude, interest, and value (Gay et al., 2012). Attitude is closely linked to the current study, and several types of attitude scales exist, one of which is the *Likert* scale and another, the *Rating* scale. These scales allow respondents to self-report along a continuum of choices (Gay et al., 2012). Also, the responses can be recorded numerically, thus allowing for imperial analysis.

The survey instrument was the *Persistence* and *Academic Success Inventory* (PASI) questionnaire. The five-point *Likert* scale of measurement was used for the indication of agreement or disagreement, ranging from "*Strongly Disagree"* as a 1-point to "*Strongly Agree"* as a 5-point response. Participants responded to the various item statements using this scale. The *Rating* scale of measurement was used for the numerical indication of preference, ranging from "*Never"* as a 1-point to "*Always"* as a 5-point response. Just as in the Likert scale of measurement, participants responded to the various item statements pertaining to this scale. For social activities, the *Rating* scale of measurement was in time frames of hours-per-week, ranging from "*Less than 1"* as a 1-point to "*More than 15"* as a 5-point response. Furthermore, three questions were reverse-scored with "*Less than 1"* as a 5-point to "*More than 15"* as a 1-point response. Comments from participants were used to analyze results and give recommendations for future research. Both the nominal and the ordinal scales of measurements were used for the general demographics.

Each item on the instrument addressed some demographic or institutional factor in the study. Recall that the two independent variables are demographic factors (*gender and ethnicity*) and institutional factors (*campus interactions, school climate, support services,* and *social activities*). Furthermore, there are two dependent variables - *persistence* and *academic success* that are the outcomes or constructs in the study. Based on this, items were grouped together under each construct.

Validity and Reliability of the Instrument

In this study, effort was made to ensure that the measuring instrument was valid and reliable. There are various ways to determine the validity and reliability of the instrument. However, only those that are relevant to this study were adopted, computed, and discussed.

Validity of the Instrument

Validity refers to the degree to which a test measures what it is supposed to measure and, consequently, permits appropriate interpretation of scores (Gay et al., 2012). Validity also means the accuracy, usefulness, appropriateness, or meaningfulness of an instrument. According to Bui (2009), validity refers to the extent to which the instrument measures what it was intended to measure. Gay et al. (2012) further state that validity of an instrument is specific for a particular purpose and the particular group in a study. Thus, validity of the aforementioned PASI survey instrument was discussed only for the purpose of this particular study.

The validity of the PASI instrument for this study was determined through content validity. Content validity is the degree to which a test measures an intended content area (Gay et al., 2012), and requires both item validity and sampling validity. According to the authors, item validity is concerned with whether the test items are relevant to the measurement of the intended content area, and sampling validity is concerned with how well the test samples the total content area being tested. Content validity is determined by expert judgment or by a group of experts in that particular field (Gay et al., 2012). Thus,

prior to administering the survey, the instrument was given to a group of academic and research experts to determine the validity of its contents, and to ascertain that it is meeting standards and measuring the *persistence* and *academic success* of the international students based on these selected demographic and institutional factors.

Content validity is of particular importance for this instrument because it is an original instrument. When the contents of the instrument were modified after the pilot test, the group of experts approved the instrument. It was then that the survey was administered. Thus, the overall validity of the study results was not threatened.

Reliability of the Instrument

Reliability is the degree to which a test consistently measures whatever it is measuring (Gay et al., 2012). Kerlinger and Lee (2000) acknowledge that reliability is the lack of distortion or the precession of a measuring instrument. The authors note that a reliable instrument is dependable, stable, consistent, and predictable. According to Bui (2009), reliability refers to the extent to which an instrument consistently measures what it was intended to measure. To determine the reliability of the instrument in this study, a reliability coefficient was computed for the items on the instrument.

For internal consistency reliability, the Cronbach's alpha reliability approach was used. Cronbach's alpha estimates internal consistency reliability by determining how all items on a test relate to all the other test items, as well as to the test as a whole (Gay et al., 2012). According to the authors, internal consistency results when all the items on the test or instrument are measuring similar things. Lund and Lund (2018) state that

Cronbach's alpha is the most commonly used measure of internal consistency or reliability, especially when there are multiple Likert questions in a survey or questionnaire. Cronbach's alpha was chosen for this study because the items representing the institutional factors have different Rating scales.

Before the study began, the total survey was administered to a group of 30 participants as a pilot study. Each participant's score for an item was computed automatically through the survey collection database. Results were then exported to the Statistical Package for the Social Sciences (SPSS) software version 19. Using the SPSS Statistics, the reliability analysis was run for items on the survey instrument having Rating scales. These were the responses to the items regarding *institutional* factors which were used in measuring the students' *persistence* and *academic success*.

On the *Persistence* and *Academic Success Inventory* (PASI) survey instrument, each institutional factor had five sub-item responses. Cronbach's alpha reliability analysis was run on each set of sub-items for the factor separately to determine the reliability of that scale. The value for each sub-item was summated for the 30 participants, after which the sets of scores were correlated under the SPSS statistical inter-item area. The reliability statistics table provided the actual Cronbach's alpha value for the particular scale (Lund & Lund, 2018). For campus interactions, Cronbach's alpha is 0.768; for school climate, it is 0.816; for support services, 0.790; and 0.663 for social activities. Though the alpha coefficient for social activities is less than 0.70, the researcher tested it for significance at the .001 level and it was found to be significant. Thus, the collective results determined that the instrument was reliable.

Data Collection Procedures

Data collection employed for this study was through the use of the questionnaire. After Institutional Review Board (IRB) approval for the study, the PASI questionnaire was administered to the participants electronically through Survey Monkey. The survey link was sent to these students through the school's International Students Office. Face-to-face recruitment of study participants and delivery of the PASI survey instrument was available on campus. Such times were during the international students' Curricular Practical Training (CPT) and Optional Practical Training (OPT) workshops, international students' orientation, and their multicultural days, as well as days that the International Students Office had major activities and required students to be available on campus.

As mentioned previously, the questionnaire had three sections (demographic information, institutional information, evaluation and expectations). When scoring the responses, the items under each section were grouped together under constructs or categories. For instance, values for all 5 items under *"Campus Interactions"* for each student were summated. The same was done for the other levels of institutional factors. Thus, each factor was rated by the 346 participants in the study. The rating score for each institutional factor selected was given as the total score for that factor divided by the number of sub-items it has.

Pilot Test/Field Study

When a project is new and an instrument is an original creation, there has to be a trial of some sort to determine if the instrument will work for the project. This is done in

two stages, through pilot testing and field testing (Second Language Testing Inc. [SLTI], 2018). The University of Phoenix (2015) acknowledges that field tests and pilot studies are used as strategies to check the soundness or "validity" of a data collection instrument. According to SLTI (2018), pilot testing is a small-scale trial, where a few participants take the test and then comment on the techniques of the test. The next stage is the field test. SLTI (2018) states that the primary purpose of a field test is to determine if the instrument is valid and reliable before it is administered to the participants.

As a quantitative study, the PASI survey instrument was used to collect data. To test the survey instrument, the validity and reliability of the survey items were statistically analyzed through the pilot study (University of Phoenix, 2015). Validity and reliability of the instrument have already been discussed in the previous sections. The final instrument was produced after the pilot and field tests were conducted and required modifications were made. For this current study, the pilot and field tests were combined, with 30 international students as participants.

As part of the pilot study, a group of experts were given the instrument to review. Notably, they looked at the guidelines and instructions on the instrument to make sure they are understandable. The research setting or venue where the data collection would take place was not fixed as the instrument was both electronic and face-to-face. Thus, the participants could take the survey anywhere. To take care of the face-to-face settings, the experts assured that the classrooms and other venues, as well as any other infrastructure that could hinder the data collection process, were appropriate and adequate.

Though this study involved human subjects, no IRB review was needed for the field-testing of the items on the questionnaire. This is because the researcher's aim was only to ensure that the questionnaire items were written appropriately for data collection based on experts' review of the questions (University of Phoenix, 2015).

Independent and Dependent Variables

In this quantitative study, important variables were identified, as their interaction determined the outcome of the research after it took place. The independent variable in any study is that which the researcher manipulates or observes (Gay et al., 2012). This variable influences or explains the phenomenon under investigation. This phenomenon is the dependent variable, identified as the outcome or criterion variable (Gay et al., 2012). In this current study, the independent variables were not manipulated. Instead, they were observed responses from the participants to the items on the survey instrument. The independent variables in this study were selected demographic and institutional factors influencing the international students' *persistence* and *academic success*. The dependent variables were, ultimately, outcomes of their academic progress.

Demographic and institutional factors were observed for variability and influence. The demographic factors were categorized as *gender* and *ethnicity*. The institutional factors were categorized as *campus interactions, school climate, support services*, and *social activities*. The phenomena in the study were the students' *persistence* and *academic success*. As one of the outcome variables studied, *persistence* was assessed by the international student's decision to return to the institution the following semester.

Academic success, the second outcome variable, was assessed by the students' cumulative (overall) institutional GPA at the time of data collection. The flowchart below (Figure 2) highlights the independent and dependent variables in the study, thus showing the relationship between the variables.

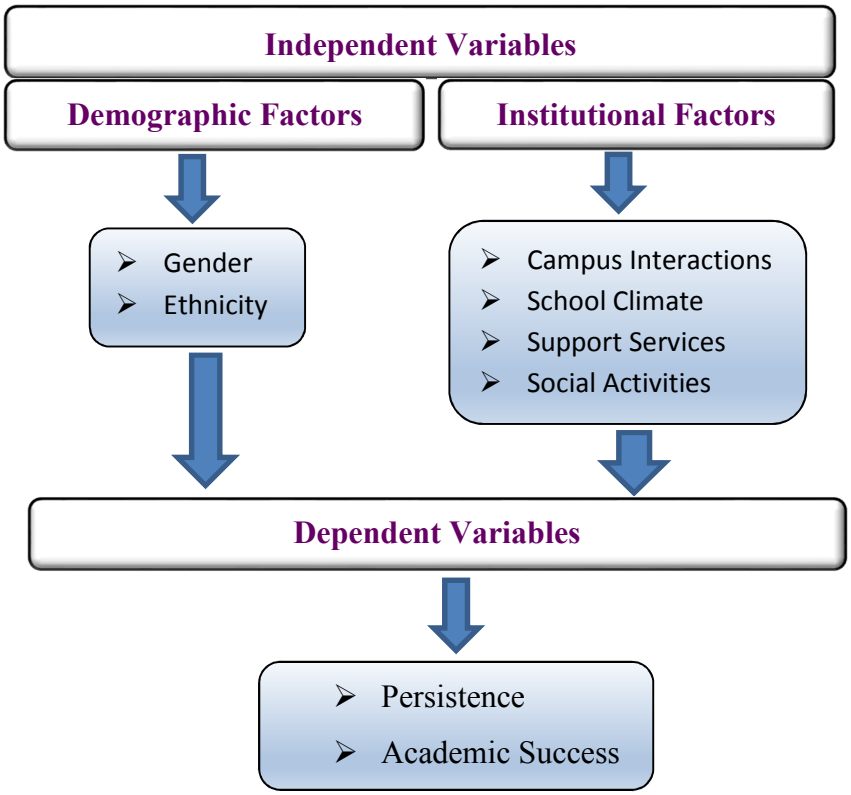


Figure 2. Independent and Dependent Variables in the Study

Null Hypotheses

Before the study took place, there was the need to establish research questions and determine the hypotheses to be tested. A hypothesis is a tentative or conjectural statement about the relationship between two or more variables (Kerlinger & Lee, 2000). It is more than an educated guess. This research investigated the connection between the

independent variable and the dependent variable, more especially, the correlation between the selected independent variables and the dependent variables. However, the most important concept is the null hypothesis. According to Gay et al. (2012), the null hypothesis states that there is no significant relation or difference between variables. It is the already existing theory that the study tested. Results would confirm or negate this theory. In this study, the two dependent variables were assessed on two levels (the independent variables). Thus, there were two null hypotheses for each of the independent variables, making a total of four null hypotheses. The null hypotheses (H_{01}, H_{02}, H_{03}, and H_{04}) are stated below:

H_{01}: There is no statistically significant relationship between the demographic factors (gender and ethnicity) and persistence among international students.

H_{02}: There is no statistically significant relationship between the institutional factors (campus interactions, school climate, support services, and social activities) and persistence among international students.

H_{03}: There no statistically significant relationship between the demographic factors (gender and ethnicity) and academic success among international students.

H_{04}: There is no statistically significant relationship between the institutional factors (campus interactions, school climate, support services, and social activities) and academic success among international students.

Statistical Analyses

The phenomena under investigation are the *persistence* and *academic success* of international students. The researcher assessed *persistence* based on whether the student responded returning to the same institution the following semester. The researcher further assessed *academic success* through the responses to the institutional items on the survey instrument. In this study, two independent variables predicted the phenomena. The first was the demographic characteristics (*gender* and *ethnicity*) of the international students. Gender is categorical, and study participants fell into male and female categories. Ethnicity was also categorical. The second independent variable was the selected institutional factors that the students encountered. These factors were broken down into *campus interactions, school climate, support services,* and *social activities*.

This structure of the study invariably determined the type of statistical analysis carried out for the research. Two independent variables meant that there were two predictors (*demographic* and *institutional* factors), while two dependent variables indicated two outcomes (*persistence* and *academic success*). The nature of these dependent variables defined the research and prompted two choices for the statistical analyses. Thus, the study applied the *Binary Logistic Regression* and the *Standard Multiple Regression* models for statistical analyses of the data (See figures 3 and 4 for the design layouts). Applying these models, the researcher tested the four null hypotheses generated (see section on null hypotheses).

Logistic Regression Design Layout

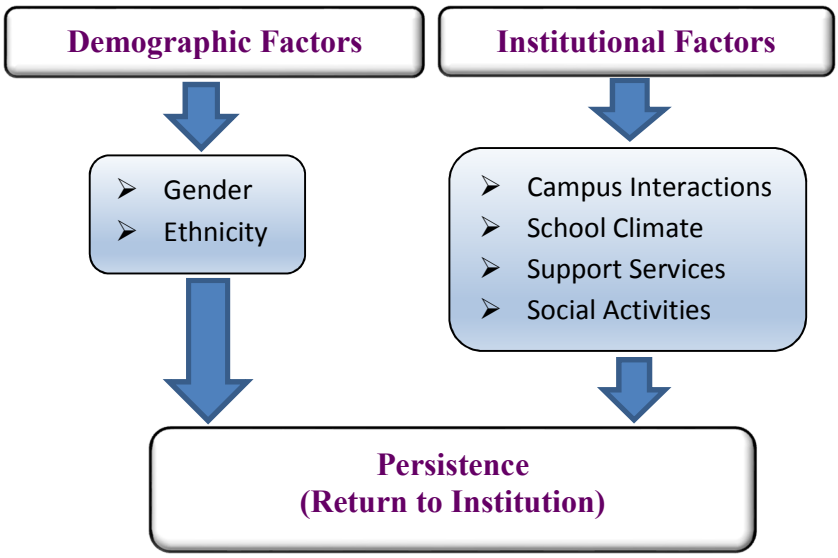


Figure 3. Predictive Power of Demographic and Institutional Factors on the Persistence of International Students

Logistic Regression

Researchers use the Logistic Regression model when the outcome variable is binary or dichotomous (Pandey, 2017). According to Pandey (2017), it means the dependent variable takes only two possible values, such as "yes" or "no." The author notes that this model has several advantages.

1) The predictor variables or the independent variables do not need to have a normal distribution (bell shaped curve), be linearly related, or have equal variances (homogeneity of variance) within each group.

2) There is also no need for the error terms (or residuals) to be normally distributed.

3) Homoscedasticity is not required, and extreme values or outliers in the predictors that are interval/ratio do not influence the outcome.

Pandey (2017) also stated that the independent variables could either be categorical variables or have numerical values. Also, with a large sample size, extraneous variables could be controlled. Logistic Regression was applied to predicting student *persistence* in this study.

Multiple Regression Design Layout

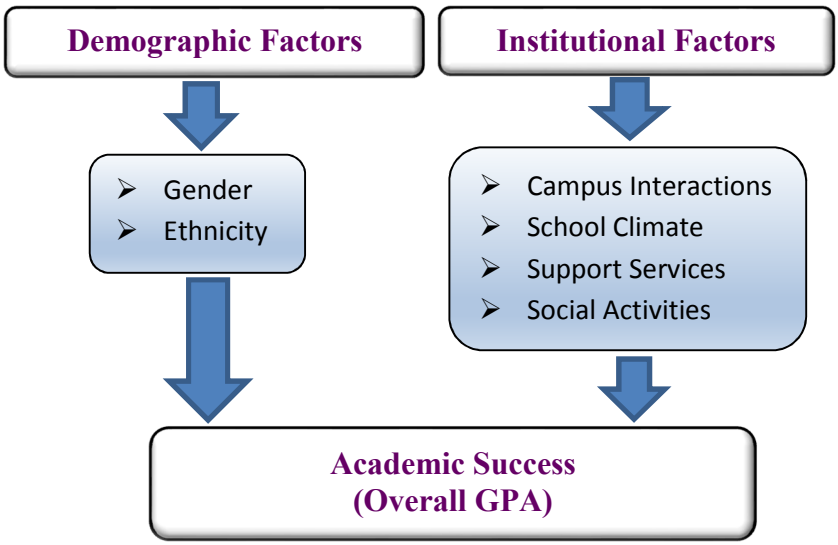


Figure 4. Predictive Power of Demographic and Institutional Factors on the Academic Success of International Students

Multiple Regression

Similarly, Multiple Regression models were used in prediction studies where the combination of independent variables predicts the dependent variable. According to Gravetter and Wallnau (2013), multiple regression is the process of using several predictor variables to obtain a more accurate prediction of the outcome variable. Kerlinger and Lee (2000) further stated that multiple regression enables the researcher to learn about how relative the influences of the independent variables are on the dependent variable. Using the Standard Multiple Regression model, the regression equation is:

$$Y= a + b_1X_1 + b_2X_2$$

where Y is the outcome variable, X_1 and X_2 are the predictor variables in the study, b is the slope of the regression line, and "a" is the Y-intercept. Extraneous variables can be controlled by employing the model as a multiple leaner regression model, as if each predictor was analyzed independently (Mertler & Vannatta, 2013). By applying this equation for the analysis, this present study adopted the Multiple Regression method to predict the *academic success* of the international students.

Furthermore, the researcher tested the multiple regression equation for statistical significance. Though more complex, Kerlinger and Lee (2000) stated that the tests of statistical significance in multiple regression are based on the idea of comparing variances (or mean squares) as in the analysis of variance. A linear regression model was not applied to this study because there was more than one predictor variable (the independent variable) in the study. Considering the probability of a type one or a type two error, statistical assumptions were adhered to.

Evaluation of Statistical Assumptions

With any model used for statistical analysis come assumptions. Both the Logistic and Multiple Regression models dealt with these assumptions. Statistical assumptions are important, as data that do not conform to these assumptions will result in biased analysis (Mertler & Vannatta, 2013). Each model had assumptions based on the nature of the variables in the study. Following the guidelines applicable to the assumptions, the researcher was able to analyze and report the findings accurately.

Logistic Regression Assumptions

1) The observations must be independent of each other and not come from repeated measurements or matched data (Lani, 2018). When each response in the questionnaire is added, the total value of all items will be determined. Thus, the individual scores were used to establish independence of observations.
2) Logistic regression requires the predictors or independent variables in the study not to be highly correlated with each other. Kassambara (2017) states that the model assumes no high intercorrelations (i.e. multicollinearity) among the predictors.
3) The dependent or outcome variable must be dichotomous, such as “yes” or “no.” This variable must also not be measured on an interval or ratio scale. Kassambara (2017) states that the outcome can also be a binary variable, like “1” or “0.”

4) Analysis usually requires a large sample size (Lani, 2018). The author states the general guideline as having a minimum of 10 cases for each independent variable in the study.

Multiple Regression Assumptions

1) Regression assumes that variables in the study have normal distribution (Osborne & Waters, 2002). This means that the variables form a bell-shaped curve.
2) The relationship between the independent and dependent variable(s) must be linear in nature. Mertler and Vannatta (2013) indicate that this is a straight line relationship.
3) Homoscedasticity is assumed. Mertler and Vannatta (2013) state that, just like in homogeneity of variances, homoscedasticity is the assumption that the variability in a continuous dependent variable is the same at all levels of the independent variable or discrete grouping. According to Lane (n.d.), this assumption means that the variance around the regression line is the same for all values of the predictor variable (X).
4) There has to be variance in all predictors. Strand, Cadwallader, and Firth (2011) emphasize the importance of varying explanatory variables even though they may be ordinal, nominal, or continuous. Strand et al. (2011) note that each variable must have at least a small range of values even if there are only two categorical possibilities.

5) Variables (particularly independent variables) are measured without error (reliably) (Osborne & Waters, 2002, Mertler & Vannatta, 2013).

Summary

In this chapter, the detailed research process was discussed. It was important to indicate all the steps involved, to be certain that the study was properly carried out. Also, the different requirements that enabled the researcher to successfully execute the study were highlighted. With the detailed research methodology, the researcher evaluated and addressed potential issues that would hinder the authenticity of the results and research findings. These issues were taken care of at the beginning of the research process, enabling the researcher to have a high degree of confidence with the outcome of the study, the results, and the findings thereof.

CHAPTER 4

DATA RESULTS AND ANALYSES

In the study, data collected were analyzed descriptively and inferentially using the SPSS software. Expectations were that the responses to the survey questions would help in understanding the purpose of this study, which was to examine the predictability of selected demographic and institutional factors on the *persistence* and *academic success* of international students. More specifically, this study was concerned with the predictive power of demographic factors (*gender* and *ethnicity*) and institutional factors (*campus interactions, school climate, support services,* and *social activities*) on the *persistence* and *academic success* of international students.

Answers to the following questions were sought in the study:

1) Do demographic factors (gender and ethnicity) have any predictive power with regard to persistence among international students?
2) Do institutional factors (campus interactions, school climate, support services, and social activities) have any predictive power with regard to persistence among international students?
3) Do demographic factors (gender and ethnicity) have any predictive power with regard to academic success among international students?
4) Do institutional factors (campus interactions, school climate, support services, and social activities) have any predictive power with regard to academic success among international students?

The sample size for this empirical study consisted of a total of three hundred and forty-six (346) international students enrolled in one of the Historically Black Colleges and Universities (HBCUs) located in an urban metropolitan area in the southern region of the United States.

For analysis of data results, this chapter was divided into four major sections. The first section, based on descriptive statistics, consisted of the demographic makeup of the study participants and the breakdown of their characteristics by gender and ethnicity. In the second section, the means and standard deviations of the variables used in the regression models were compiled. The third section featured the correlation between the independent and the dependent variables. The fourth section was where the null hypotheses formulated for this study were tested. The binary Logistic Regression was used to test the null hypotheses on student *persistence* and the standard Multiple Regression was used to test the null hypotheses regrading *academic success.*

Demographic Characteristics of Study Participants

A total of three hundred and forty-six (346) international students participated in the study. These were students from diverse ethnic backgrounds with varying demographic characteristics. Thus, they were descriptively categorized by gender and ethnicity.

Gender

There were 172 (49.7%) male students and 174 (50.3%) female students in the study (see Table 1 for findings).

Table 1

Frequency Distribution of Study Participants by Gender

Variable	Number	Percent
Male	172	49.7
Female	174	50.3
Total	346	100.0

Ethnicity

The ethnic groups represented by the participants in this study were categorized into four subgroups. There were 172 (49.7%) students who indicated that they were of African descent, and 72 (20.8%) who reported their ethnicity as Asian origin. Furthermore, 76 (22.0%) participants identified their ethnicity as Middle Eastern, while 26 (7.5%) participants whose ethnicity was not expressed as one of the major ethnic groups indicated above were categorized as “Other” (see Table 2 for these findings).

Table 2

Frequency Distribution of Study Participants by Ethnicity

Variable	Number	Percent
African Descent	172	49.7
Asian	72	20.8
Middle Eastern	76	22.0
Other	26	7.5
Total	346	100.0

Mean and Standard Deviation Results

For the Regression analyses, the mean and standard deviation (SD) of the independent and dependent variables were compiled (see Table 3). The dependent variables were *persistence* (measured by return to institution) and *academic success* measured by *GPA*. From the study, the mean score for *persistence* was 1.16 (SD = .37) and the mean score for *academic success* was 3.38 (SD = .31). There were two independent variables, demographic and institutional factors, which were further categorized and broken down into subgroups.

With regard to the demographic factors, the variables were categorized as gender and ethnicity. Ethnicity was further sub-grouped into African Descent, Asian, Middle Eastern, and "Other" ethnicity. To better organize the data, new variables were created through dummy coding. Male was re-coded as "0" and female as "1" to make male the

reference group to compare gender in the regression models. The re-coding for ethnicity was done likewise. The ethnic group "Other" was re-coded from "4" to "0" for "Other" and "0" for any group that is not "Other." African Descent was re-coded from "1" to "1" for African Descent, and "0" for any group that is not African Descent. Similarly, Asian was re-coded from "2" to "1" for Asian, and "0" for any group that is not Asian. Furthermore, Middle Eastern was re-coded from "3" to "1" for Middle Eastern, and "0" for any group that is not Middle Eastern. With this recoding strategy, the ethnic group "Other" was made the reference group in which to compare ethnicity in the models.

Table 3

Mean and Standard Deviation Results for the Independent and Dependent Variables

Variable	**Mean**	**SD**
Academic Success (Overall GPA)	3.3836	.30905
Persistence (Return to Institution)	1.1647	.37148
Female	.5029	.50072
African Descent	.4971	.50072
Asian	.2081	.40653
Middle Eastern	.2197	.41461
Campus Interactions	3.0023	.75281
School Climate	3.2353	.75774
Support Services	3.2434	.79248
Social Activities	3.2850	.89083

Correlation Analyses between Independent and Dependent Variables

There are two independent variables in the study (*demographic* and *institutional* factors). The two dependent variables in the study are *persistence* and *academic success*. Correlation analysis was used to determine the relationship between the dependent and independent variables. The Biserial (r_b) and Point-Biserial Correlations (r_{pb}), as well as the Pearson Product-Moment Correlation (r), were used to assess the relationship between the variables utilized in the regression models (see Table 4).

Relationships between demographic factors (*gender* and *ethnicity*) and *persistence* (with category breakdowns) were moderately positive, but none had statistical significance. However, the relationship between institutional factors and *persistence* was positive for support services (r_{pb} = .155) and social activities (r_{pb} = .156). They were also found to be significant. For the relationship between demographic factors (*gender* and *ethnicity*) and *academic success*, gender was moderately positive with no significance. Ethnicity was significant and positive for those of African Descent (r_b = .112). However, for those of Middle Eastern ethnicity, the relationship was negative, though significant (r_b = -.132). Furthermore, there was the relationship with institutional factors and *academic success*. Here, campus interactions had a significant but negative relationship (r = -.120), while social activities was also significant but with a positive relationship (r = .121).

Table 4

Correlation Results between Independent and Dependent Variables in the Regression Models

Independent Variable	Dependent Variables	
	Persistence (Return to Institution)	**Academic Success (Overall GPA)**
Female	.021	.074
African Descent	.010	.112*
Asian	.003	.024
Middle Eastern	.028	-.132*
Campus Interactions	.055	-.120*
School Climate	.068	-.054
Support Services	.155**	.090
Social Activities	.156**	.121*

Note. *Significant at the 0.05 level. **Significant at the 0.01 level.

Testing the Hypotheses for Persistence

H_{01}: There is no statistically significant relationship between the demographic factors (gender and ethnicity) and persistence among international students.

The study applied a binary Logistic Regression technique to determine the relationship and predictive power of the variables gender and ethnicity on *persistence* among international students. After data screening, unusable cases and outliers were

removed from the analysis. In the regression results, the chi square for the Hosmer and Lemeshow goodness-of-fit test produced a non-significance p-value (p =.636) with a large log-likelihood. Bartlett (2014) states that a small p-value indicates a poor fit. Field (2009) concurs, indicating that a p-value of 1 means that the model predicts the outcome variable perfectly.

Though the model in the current study predicted the outcome variable to a good extent, the large log-likelihood value indicates that the model did not fit the data statistically, as there were more unexplained than explained observations (Field, 2009). Here, the two predictors (*gender* and *ethnicity*) were not statistically reliable in distinguishing between international students who would persist and return to the institution the following semester (-2 Log Likelihood = 307.418, χ^2 (5) = 3.419, p = .636) and those who would not (see Table 5).

Results of Nagelkerke's R Square test for effect size indicated the same, where the variables gender and ethnicity together accounted for only 1.1% of the variance in *persistence* by the model (see Table 5). However, prediction of *persistence* was very accurate in terms of correctly classifying those international students who would not persist or return to the institution the following semester (0%). Furthermore, the overall model correctly classified 83.5% of international students as returning to the institution the following semester (see Table 6).

Table 5

Overall Model Fit Results Regarding the Relationship between Demographic Factors and Persistence

Model	Chi Square	df	p
Final	3.419	5	.636

-2 Log Likelihood = 307.418; Nagelkerke's R Square = .011.

Table 6

Classification Table Results Regarding Demographic Factors and Persistence

Persistence	*Not Return to Institution*	*Return to Institution*	*Percentage Correct*
Not Return to Institution	0	57	.0
Return to Institution	0	289	100.0

Overall Correct 83.5%

The Wald Statistic, a measure of significance in logistic regression (Mertler & Vannatta, 2013), together with their associated significance value, was used to test the significance of each independent predictor on *persistence* among international students (see Table 7). Study results show that each predictor made no significant contribution to the outcome. For gender, the Wald test displays the value for female as ($z = .239$, $p = .625$). For ethnicity, the test displays the value for African Descent as ($z = 1.342$, $p =$

.247) and Asian as (z = 1.126, p = .289), while for Middle Eastern the value is (z = 1.675, p =.196). Here, the Wald test significance values indicated that the variables gender and ethnicity were not significant predictors of *persistence*.

Finally, the effect that the independent variable has on the dependent variable is the unstandardized regression coefficient (B). Mertler and Vannatta (2013) state that for a dichotomous outcome, this effect is expressed as an odds ratio (Exp(B) or e^{β}). The odds ratio for gender displays female as (e^{β} = 1.160). For ethnicity, the odds ratio shows African Descent as (e^{β} = 2.422), Asian as (e^{β} = 2.343), and Middle Eastern as (e^{β} = 2.803). Even though there was no statistically significant relationship between African Descent, Asian, Middle Eastern, and *persistence*, if an international student was from one of these ethnic groups, he or she would be twice more likely to return to the institution.

Table 7

Regression Coefficients Regarding the Relationship between Gender, Ethnicity, and Persistence

Variable	B	SE	Wald	df	p	Exp(B)
Female	.148	.303	.239	1	.625	1.160
African Descent	.885	.764	1.342	1	.247	2.422
Asian	.852	.803	1.126	1	.289	2.343
Middle Eastern	1.031	.797	1.675	1	.196	2.803
Constant	-2.561	.753	11.554	1	.001	.077

-2 Log Likelihood = 307.418; Nagelkerke's R Square = .011, χ^2 (5) = 3.419, p = .636.

H_{02}: There is no statistically significant relationship between the institutional factors (campus interactions, school climate, support services, and social activities) and persistence among international students.

The Logistic Regression technique was used here to determine the relationship and predictive power of the variables campus interactions, school climate, support services, and social activities on *persistence* among international students. The non-significant p-value (p = .409) produced by the chi square for the Hosmer and Lemeshow test indicates that the model did not predict the outcome variable well. Similarly, with a large log-likelihood, regression results did not indicate a model fit. The overall model of the four predictors (campus interactions, school climate, support services, and social

activities) were not statistically reliable in distinguishing between international students who would be persistent and return to the institution the following semester (-2 Log Likelihood = 292.370, χ^2 (8) = 8.253, p = .409) and those who would not (see Table 8).

Table 8

Overall Model Fit Results Regarding the Relationship between Institutional Factors and Persistence

Model	Chi Square	df	p
Final	8.253	8	.409

-2 Log Likelihood = 292.370; Nagelkerke's R Square = .082.

As in the results for gender and ethnicity, Nagelkerke's R Square test for effect size showed that the variables campus interactions, school climate, support services, and social activities all together accounted for 8.2% of the variance in *persistence* by the model (see Table 8). Nonetheless, the model correctly classified the *persistence* of those international students who would not return to the institution the following semester (0%). In addition, the overall model accurately classified 83.5% of international students returning to the institution the following semester (see Table 9).

Table 9

Classification Table Results Regarding Institutional Factors and Persistence

Persistence	*Not Return to Institution*	*Return to Institution*	*Percentage Correct*
Not Return to Institution	0	57	.0
Return to Institution	0	289	100.0

Overall Correct 83.5%

Moreover, the Wald test revealed that the variables campus interactions (z = .007, p = .932) and school climate (z = .428, p = .513) were not significant predictors of *persistence*. Furthermore, the test revealed that support services (z = 7.004, p = .008) and social activities (z = 7.550, p = .006) were significant predictors of *persistence*. Finally, the more international students had access to support services (e^{β} = 1.682), and the more they participated in social activities (e^{β} = 1.621), they were twice more likely to persist than those with lower access to support services and those with lower participation in social activities (see Table 10).

Table 10

Regression Coefficients Regarding the Relationship between Institutional Factors and Persistence

Variable	B	SE	Wald	df	p	Exp(B)
Campus Interactions	.018	.216	.007	1	.932	1.019
School Climate	.142	.217	.428	1	.513	1.153
Support Services	.520	.196	7.004	1	.008**	1.682
Social Activities	.483	.176	7.550	1	.006**	1.621
Constant	-5.539	1.134	23.867	1	.000	.004

Note. -2 Log Likelihood = 292.370; Nagelkerke's R Square = .082, χ^2 (8) = 8.253, p = .409. **Significant at the .01 level.

Testing the Hypotheses for Academic Success

H_{03}: There is no statistically significant relationship between the demographic factors (gender and ethnicity) and academic success among international students.

The standard Multiple Regression analysis was employed in this study to determine the relationship and predictive power of gender and ethnicity on the *academic success* of international students. Data screening led to the elimination of unusable cases and outliers from the analysis. The regression results were generated with the variables of gender and ethnicity categorized to specify gender and the various ethnic makeup of the participants. As shown in Table 11, the regression model displayed a multiple correlation

(R) of .155. The predictors in the model accounted for only 2.4% (R^2_{adj} = .013) of variance in *academic success*.

According to Mertler and Vannatta (2013), the F test examines the degrees to which the predictors and the outcome are linear. The F test indicated that no linear relationship was found to exist between the predictors of gender and ethnicity and the outcome of *academic success* [F (4, 341) = 2.092, *p* = .081]. Mertler & Vannatta (2013) further state that the unstandardized regression coefficient (*B*) designates the change in the value of the dependent variable with a unit increase in each predictor in the model, and is expressed as the slope weight for that predictor.

Table 11

Regression Coefficients Regarding the Relationship between Gender, Ethnicity, and Academic Success

	B	SE	Beta (β)	t	p
(Constant)	3.323	.063		53.073	.000
Female	.028	.034	.046	.830	.407
African Descent	.080	.065	.130	1.238	.217
Asian	.056	.070	.074	.795	.427
Middle Eastern	-.024	.070	-.032	-.342	.733

Note. R = .155; R^2 = .024; Adjusted R^2 = .013; F = 2.092; df = 4, 95; *p* = .081.

Reviewing the regression coefficient (*B*) for gender in the model, females displayed a coefficient of (*B* = .028). However, gender did not significantly contribute to the model [t (341) = .830, *p* = .407]. Regarding ethnicity, African Descent presented a coefficient of (*B* = .080) with no significant contribution to the model [t (341) = 1.238, *p* = .217], while Asians exhibited a regression coefficient of (*B* = .056), also with no significant contribution to the model [t (341) = .795, *p* = .427]. Furthermore, Middle Eastern revealed a coefficient of (*B* = -.024) with no significant contribution to the model [t (341) = -.342, *p* = .733].

As indicated, none of the ethnic subgroups contributed significantly to the model, nor were they independent predictors of *academic success* (see Table 11 for the respective t-scores and *p*-values).

H_{04}: There is no statistically significant relationship between the institutional factors (campus interactions, school climate, support services, and social activities) and academic success among international students.

Just as that for gender and ethnicity, the Multiple Regression procedure was employed here to determine if a relationship existed between the institutional factors (campus interactions, school climate, support services, and social activities) and *academic success* among international students. Results from the F test highlight that a linear relationship was found to exist between the institutional factors (campus interactions, school climate, support services, and social activities) and the outcome of *academic success* [F (4, 341) = 3.998, *p* = .003]. Thus, regression results produced an

overall model that was successful in predicting *academic success*. In Table 12, however, the regression model displayed a multiple correlation (R) of .212, with the predictors accounting for only 4.5% (R^2_{adj} = .034) of variance in *academic success*.

Table 12

Regression Coefficients Regarding the Relationship between Institutional Factors and Academic Success

	B	SE	Beta (β)	t	p
(Constant)	3.291	.111		29.724	.000
Campus Interactions	-.060	.025	-.145	-2.420	.016*
School Climate	-.009	.024	-.022	-.374	.708
Support Services	.050	.021	.127	2.322	.021*
Social Activities	.042	.019	.122	2.285	.023*

Note. R = .212; R^2 = .045; Adjusted R^2 = .034; F = 3.998; df = 4, 95; *p* = .003**
*Significant at the .05 level. **Significant at the .01 level.

Reviewing the slope weights (*B*) for the institutional factors, it was seen that international students who engaged more in campus interactions had lower *academic success* than those who engaged less (*B* = -.060). Here campus interactions negatively influenced *academic success* [t (341) = -2.420, *p* = .016]. Results show that school climate negatively impacted the *academic success* of international students (*B* = -.009) though it did not contribute significantly to the model [t (341) = -.374, *p* = .708]. It was seen that international students who had more access to support services had higher

academic success than those who had lower access (*B* = .050). Thus, support services positively affected *academic success* [t (341) = 2.322, *p* = .021]. Additionally, international students who participated more in social activities had higher *academic success* than those with lower participation (*B* = .042). Here social activities positively influenced *academic success* [t (341) = 2.285, *p* = .023].

Furthermore, it was observed that campus interactions, support services, and social activities contributed significantly to the model, and were all independent predictors of *academic success* (see Table 12 for the respective t-scores and *p*-values).

Summary of the Hypotheses Tests

Four hypotheses were formulated and tested in this study. They were all tested to identify if a relationship existed between the independent and dependent variables. For H_{01} and H_{02}, the hypotheses were tested for predictive relationship between the independent variables and the dependent variable of *persistence*. Both Hypotheses were found to be non-significant, though in hypothesis H_{02}, support services and social activities were independently significant. Hypotheses H_{03} and H_{04} were tested for predictive relationship between the independent variables and the dependent variable of *academic success.* Out of the two, hypothesis H_{04} was found to be significant.

In the study, the demographic factors of gender and ethnicity together in both regression models were not reliable in predicting *persistence* and *academic success* among international students. The institutional factors combined in the Logistic Regression model did not predict *persistence.* However, these factors collectively

predicted *academic success* among international students in the Multiple Regression model.

Looking at the Logistic Regression model, support services and social activities were independent predictors of *persistence*. Furthermore, in the Multiple Regression model, campus interactions, support services, and social activities independently predicted *academic success* among international students (See Table 13 for the summary of the hypotheses tested in the study).

Table 13

Summary Table for the Hypotheses Tested in the Study

Hypothesis	Chi Square	R	R^2	df	p	Conclusion
H_{01}	3.419	-	.011	5	.636	Non-Significant
H_{02}	8.253	-	.082	8	.409	Non-Significant
H_{03}	-	.155	.024	4	.081	Non-Significant
H_{04}	-	.212	.045	4	.003**	Significant

Note. **Significant at the .01 level. [a]*p-value* for H_{01} and H_{02} are for goodness of fit. [b]*p-value* for H_{03} and H_{04} are for linearity. Conclusions are based on the results of the hypotheses tested.

CHAPTER 5

SUMMARY, RECOMMENDATIONS, AND CONCLUSIONS

Summary

The purpose of this study was to examine the predictive power of selected demographic and institutional factors on the *persistence* and *academic success* of international students enrolled in an HBCU in the southern region of the country. Of concern in this study was the predictive power of the selected demographic factors (*gender* and *ethnicity*) on the *persistence* (return to institution) and *academic success* (GPA) of international students. The study was also concerned with the predictive power of selected institutional factors (*campus interactions*, *school climate*, *support services*, and *social activities*) on the *persistence* and *academic success* of international students.

For data collection, the PASI questionnaire was administered to the international students enrolled in the institution. Three hundred and forty-six (346) participants took part in the study. The binary Logistic Regression model was used in the study to assess their *persistence*. The standard Multiple Regression procedure was used to determine their *academic success.* In this investigation, the following null hypotheses were tested at the .05 level of significance or better.

H_{01}: There is no statistically significant relationship between the demographic factors (gender and ethnicity) and persistence among international students.

H_{02}: There is no statistically significant relationship between the institutional factors (campus interactions, school climate, support services, and social activities) and persistence among international students.

H_{03}: There is no statistically significant relationship between the demographic factors (gender and ethnicity) and academic success among international students.

H_{04}: There is no statistically significant relationship between the institutional factors (campus interactions, school climate, support services, and social activities) and academic success among international students.

Study Findings

Several findings regarding international students emerged from the results of this study. Identified were the following:

1) There was no statistically significant relationship between the demographic factors (gender and ethnicity) and *persistence* among international students.
2) Gender and ethnicity together accounted for only 1.1% of the variance in *persistence*.
3) Neither gender nor ethnicity was an independent predictor of *persistence* among the international students.

4) No statistically significant relationship existed between the institutional factors (campus interactions, school climate, support services, and social activities), and *persistence* among international students.

5) Campus interactions, school climate, support services, and social activities combined accounted for 8.2% of the variance in *persistence*.

6) Support services and social activities independently were significant predictors of *persistence*.

7) A statistically significant relationship did not exist between gender, ethnicity, and *academic success* among the international students.

8) Gender and ethnicity together accounted for only 2.4% of the variance in *academic success*.

9) Gender and ethnicity were not independent predictors of *academic success* among the international students.

10) No linear relationship was found to exist between gender, ethnicity, and *academic success* among international students.

11) Campus interactions, school climate, support services, and social activities combined had a statistically significant relationship with *academic success* among the international students.

12) A linear relationship was found to exist between campus interactions, school climate, support services, and social activities, and *academic success*.

13) Campus interactions, school climate, support services, and social activities combined, accounted for just 4.5% of the variance in *academic success*.

14) Campus interactions, support services, and social activities were independent predictors of *academic success* among international students.

15) School climate was not an independent predictor of both *persistence* and *academic success* among the international students.

Discussion of Study Findings

Study findings were linked to the comments from participants and data collected from the survey. The key findings are hereby addressed, with discussions on each presented below:

Gender: In the results of this present study, no statistically significant relationship existed between gender and *persistence* among the international students. Also, gender was not an independent predictor of *persistence*. Notably, Mamiseishvili (2012) came to the same conclusions. The results of the chi-square and logistic regression performed by Mamiseishvili (2012) revealed that gender was not a significant predictor of *persistence* among international students in that study. Similarly, in the study by Sanchez and Smith (2017), the authors investigated the predictive power of gender on the *persistence* and graduation status of international students, and noted no differences with regard to gender.

Looking at the study findings by Stewart, Lim, and Kim (2015), gender also had no statistically significant effect on *persistence* or remedial status of students. Interpretation of this result could be that the remedial process in the schools had equal impact on *persistence* for male and female students alike. Stewart et al., (2015) further cited similar findings where gender played no significant role in the *persistence* of students.

Further findings in this present study indicate no statistically significant relationship between gender and *academic success*. These findings match that of Sanchez and Smith (2017), where gender differences were nonexistent with regard to *academic success*. Sanchez and Smith (2017), as well as Young and Snead (2017), noted that international students experienced challenges whether they were male or female, though they reported gender-related issues. In their study, Young and Snead (2017) noted research findings which indicated that, regardless of gender, participants experienced challenges and barriers to success as international students. They faced challenges just for the fact that they were international students.

Relating these findings to the current study is particularly understandable. A plausible explanation for the study results may be dependent on several factors. This is because studies have assessed *persistence* over the years and have cited varying results regarding the predictive power of gender (Stewart et al., 2015). Even with these results, it was determined that differences based on gender were rare among international students, as other more significant factors predicted *persistence* among this group.

Ethnicity: In the results of this present study, no statistically significant relationship existed between ethnicity and *persistence* among the international students. Also, ethnicity was not an independent predictor of *persistence*. Results and conclusions were consistent with those of Mamiseishvili (2012) and Navarro et al. (2014). Ethnicity did not impact *persistence* in the study by Navarro et al. (2014). Furthermore, study results by Mamiseishvili (2012) revealed the same outcome for ethnicity, particularly for Black, Hispanic, and Asian respectively. However, for *academic success*, Mamiseishvili (2012) found ethnicity to be significant.

On the other hand, the results of the current study are not supported by those of Perry et al. (2017) regarding ethnicity. Perry et al. (2017) found that the issue of language barriers was a great concern for international students. This is more so for Asian students who cannot translate their language directly to English. Sullivan (2018) concurred, stating that the impact was based on the student's level of English proficiency. These statements indicate that the ethnicity of an international student can predict *persistence* when language barriers are considered, especially as there are countries where English is not the first language.

A reasonable interpretation for study findings may be deduced from responses given by participants. Many international students indicated some form of sponsorship based on ethnicity or country of origin. These sponsorship types and terms of enrollment in the institution may be the driving factor that cause the students to persist from semester to semester. Furthermore, they have specific course load instructions as international students, and must attain *academic success* before returning home.

Campus Interactions: Study results revealed that campus interactions did not significantly predict *persistence*. However, results depicted a statistically significant relationship between campus interactions and *academic success* among international students. This institutional factor had a negative relationship. Thus, the more the international students interacted with particular people on campus, the more their *academic success* was negatively impacted. These findings are supported by those of Amirali and Bakken (2015), which show that international students have challenges when it comes to communicating with staff.

Study results by Amirali and Bakken (2015) revealed office procedures where there was really no communication with the students after notifying them of admissions and prior to their arrival, as well as no help getting to campus and starting academic life. Furthermore, Mamiseishvili (2012) attested to the importance of peer interactions, and saw validations from faculty, staff, and peers as critical for international students. Thus, the negative relationship could be the reactions to not-so-friendly encounters by the students. Perry et al. (2017) have similar views, stating that international students who made American friends adapted better.

Adding to these findings, results by Curtin et al. (2013) indicated that international students had no positive report regarding advisor support. Notably, Zhang and Dinh (2017) dispute these findings, emphasizing the significant role of academic advisors. The authors state that with constant and continued interaction with academic advisors, the learning experiences for international engineering students improved.

Unfortunately, Curtin et al. (2013) noted that, even with a non-positive report, this category of students needs these advisors more than their American colleagues.

A plausible interpretation of the study findings might be obtained from responses and various comments made by study participants. Students interacted more frequently with fellow students but indicated unfriendly faculty and staff. Some revealed that they had no need to meet with an academic advisor as they did not care about their students and are not supportive. Others felt that they were not treated as students but as "foreigners," and felt that their International Students Office could do a better job regarding communicating to them about their international status. This apparently signified negative interactions on campus.

School Climate: Based on the results of this study, school climate had a negative impacted on *academic success* among international students. Interestingly, this institutional factor was not an independent predictor of both *persistence* and *academic success* among international students. Matching the current study results of a negative relationship, Amirali and Bakken (2015) note in their study the not-so-friendly environment that the international students report. The researchers stated that these students recounted no assistance upon arrival on campus, nor while seeking accommodation. Contrary to the findings in the present study, Perry et al. (2017) recalled results of a positive classroom climate, stating activities and contributions from international students that help create diversity.

A reasonable explanation for the study results may be deduced from responses given and comments made by participants. Students indicated a not-so-conducive school climate. Though there were many positive comments, there were also comments regarding inadequate classroom learning environments, lack of study areas, and inappropriate temperature levels, as well as poor quality restrooms. Also, comments included the quest for students and staff to be friendlier to international students. Furthermore, participants stipulate that the school does not care about their international students, and they did not have enough information regarding immigration, financial sponsoring, and good ways for cultural awareness. These comments can be attested to by the negative weight (*B*) of school climate in the study findings.

Support Services: Understandably, the issue of support services is very important. This institutional factor portrayed a statistically significant relationship with *persistence* and *academic success* and was an independent predictor of such among international students in the study. Based on study findings, the odds ratio displayed that international students who had more access to support services were twice as likely to persist as their counterparts who had lower access to these services. In other findings, this institutional factor had a positive effect on the *academic success* of international students in the study, indicating that students with more access to support services have higher *academic success* than those who have lower access to such.

Results in the current study are consistent with those of Perez-Encinas and Ammigan (2016). Their study discussions showed that support services positively

influence academic achievement. Perez-Encinas and Ammigan (2016) reported that feedback from students enabled institutions to make notable changes in their student advising, customer service, and outreach, to name a few. Singaram et al. (2012) also had consistent results with regard to tutorial groups, while Dancer et al. (2015) pinpointed peer-to-peer study groups as being effective. For successful *persistence* of international students, Mamiseishvili (2012) further noted the importance of academic advising. Findings also match those in the study by Sullivan (2018). Reports indicated that the International Students Office in the institution where the study was conducted provided every resource possible for the students, ranging from welcome toiletry packets, to transport and retail shopping experiences due to the location of the campus.

A plausible explanation for current study findings may be that access to support services such as tutoring, wellness programs, and advising, within the international student community ultimately aids in *persistence* and *academic success*. Nonetheless, comments from study participants refuted this result. Students lamented the limited access time to services such as computers and printers. Others worried about higher tuition than the local students, fewer on-campus jobs, and not enough scholarships and financial assistance. Thus, weather increased or limited access to, support services in any form are always a significant predictor of *persistence* and *academic success* among international students.

Social Activities: There was a statistically significant relationship between social activities and both *persistence* and *academic success* among international students in this

present study. Interestingly, this institutional factor was also an independent predictor of both *persistence* and *academic success*. Study findings reveal that international students who participated more in social activities were twice more likely to *persist* than their colleagues who had lower participation in these activities. Further findings show that social activities had a positive influence on *academic success*, signifying that international students who participated more in these activities had higher *academic success* than their colleagues who had lower participation.

The results of this current study were compatible with those of Sullivan (2018). Study results by Sullivan (2018) indicated that international students reported campus activities as an important factor in their campus experiences. Students noted these International Students Office planned activities as their source for outings, entertainment, cultural and social events, as well as leadership opportunities. Results in the study by Mamiseishvili (2012) stated otherwise. Findings by Mamiseishvili (2012) show that social integration had a negative impact on *persistence*. The author noted that such results could be that the international students were involved in other campus activities not captured in that study.

A reasonable interpretation could be that the various social activities on the college campus helped international students with acculturation. It is also important to note that responses from the present study show that not all students were interested in the same social activities. Some were involved in activities that were for academic purposes, others for leisure, and yet others out of necessity. Whether positive or negative,

this institutional factor played a significant role in the *persistence* and *academic success* of international students.

Study Models: The results of this present study were obtained from study models. These models were used to observe how the independent (input) variables together predict the dependent (outcome) variables. In the current study, data for gender and ethnicity were not a model fit and accounted for very little variability in *persistence* (1.1%) with no significant relationship. These results are consistent with those of Navarro et al. (2014). In their study, results show that both gender and ethnicity were model fits for the data in the bidirectional model. The authors nonetheless reported that both variables did not significantly moderate *persistence*. Similarly, data for campus interactions, school climate, support services, and social activities did not fit the model, and accounted for only 8.2% of the variance in *persistence*. However, the model depicted a significant relationship with just support services and social activities.

Model results show a small variance in *academic success* from gender and ethnicity (2.4%). Also, no linear relationship was found to exist with *academic success* (p = .081), with no statistically significant predictability. Furthermore, the study depicted a linear relationship between institutional factors and *academic success* among international students (p = .003). Moreover, these factors combined accounted for very little variance in *academic success* (4.5%). However, campus interactions, support services, and social activities significantly predicted *academic success*.

With triangulation, coding, and analysis, findings in the current study were consistent with those by Young and Snead (2017). The authors reveal that international students faced challenges regardless of gender. The model developed by Amirali and Bakken (2015) was the 3Cs strategy (Connect, Communicate, and Collaborate), together with the Quality of Life survey. Current study findings were consistent with these models reflecting both a negative campus climate and no communication with international students. Investigating academic performance using Student-Led Objective Tutorial (SLOT) versus the CT group as models, findings by Sukhlecha et al. (2016) were significant and consistent with findings in the present study.

Study observations based on the models could be interpreted in various ways. However, a reasonable explanation for these present findings might be based on the requirements as an international student. The minimal variability in *persistence* and *academic success* could indicate that these international students to a great extent understand the importance of *persistence* and attaining *academic success*. Whether male or female, and despite their ethnicity or country of origin, they have the determination to achieve academic success away from their home country.

Research Implications

Individuals from all over the world constantly enroll in academic institutions in the United States. It is then reasonable to identify the challenges they face when they leave their country of origin. Thus, the findings in this study are important as they could

relate to practical actions, policy decisions, theoretical knowledge, or future research on the topic.

The significant impact of support services and social activities on the *persistence* and *academic success* of international students suggests that institutions should review their policies regarding these institutional factors. The International Students Offices in each institution should work together with administrators to ensure that support services are in place for these students when they arrive on campus. Concerning social activities, it would be commendable if institutions implemented best practices where such activities were part of the academic process. This would enable these students to understand the importance of their academic journey on society.

Though campus interactions did not have an influence on *persistence*, it impacted the *academic success* of the international students. School administrators should understand that the international students are like customers to the institution and need to be treated respectfully. Administrators should constantly update institutional policies that impact international students, as well as data collection processes and databases for student information. Readily available student information based on interactions across campus will guide them while on those campuses. These students might persist because it is necessary for them to stay in their program and they have no choice if they are being sponsored. However, their *academic success* could be impacted and referrals to the institution jeopardized.

The issue of school climate and its lack of significance with regard to *persistence* and *academic success* of international students should not be overlooked. Even with no

significance, students commented on the classroom environment, non-friendliness of people on campus, and the social and cultural atmosphere. Thus, school administrators should regularly assess the environment in their institution and offer practical solutions to improve their campus climate.

Finally, gender and ethnicity were not significant predictors of *persistence* and *academic success* among the international students. This suggests that no gender or country is favored to enroll for further studies in the institutions. Nonetheless, administrators and policymakers should look into the various programs in their schools to ensure that they are favorable to both male and female students, as well as the various cultures and ethnicities. This way, more international students can be welcomed to the institutions, bringing in more revenue to the institution as well as to the country.

Recommendations for Future Research

Study findings, participants' comments, and challenges during the research process prompted the following recommendations:

1) Having a better strategy for recruiting and retaining study participants, as well as getting them to respond willingly to all questions and statements. This will help in minimizing unusable data.
2) Focusing future studies on investigating other demographic factors such as marital status, family makeup, sponsorship type, accommodation type, and cross-cultural differences that can impact *persistence* and the *academic success* of international students.

3) Researching the predictive power of other selected institutional factors such as on-campus jobs, scholarships, and institutional assistance (financial, social, and emotional) on the *persistence* and the *academic success* of international students.

4) Comparing international students in 2-year colleges with those in 4-year colleges to see if institutional type has predictive power on *persistence* and *academic success.*

5) Exploring the best methods for school administrators to communicate with a culturally diversified student group, producing positive campus experiences and thus more enrollment referrals to the institution.

Conclusions

The study investigated the predictive power of demographic and institutional factors on the *persistence* and *academic success* of international students. The combination of selected institutional factors makes this study unique. Findings in the study pertain to the important issues that impact international students as they leave their home country to study abroad.

Results and findings from this present study show that gender and ethnicity were not collective or independent predictors. That notwithstanding, the odds ratio for each ethnic group in the study revealed that international students were twice more likely to persist were they from any of those groups. This indicates that there is the possibility of ethnicity having some form of impact if investigated with some other analysis. Support

services and social activities were the institutional factors that significantly predicted the *persistence* and *academic success* of international students. Furthermore, campus interactions independently predicted *academic success*, while *school climate* predicted neither *persistence* nor *academic success* among international students.

It is recognized that other institutional factors can be included in the regression model to determine their predictive power on the *persistence* and *academic success* of international students. School administrators must be made aware of these factors to help students better adapt to their respective programs. Regardless of the institutional factors in any academic environment, there will always be the need to evaluate their accessibility, quality, and availability on campus for international students. Also, it will be vital to assess the impact on their *persistence* and *academic success*.

The complex nature of the academic system makes it necessary to assess the standards involved in recruiting, enrolling, and retaining international students. Also, updating the standards in handling institutional factors accessible to these students can help administrators make meaningful decisions on how to assist them. This way, factors with negative predictability could be identified instantly, and action taken to minimize their ultimate influence on the *persistence* and *academic success* of these students.

In the educational system, the *persistence* and *academic success* of international students is very important. With the benefits that these individuals bring to the various institutions, school administrators should explore the challenges they face as they leave their home country to study and make effort to address them through policy decisions or practical institutional actions.

REFERENCES

Ambrósio, S., Marques, J. F., Santos, L., & Doutor, C. (2017). Higher education institutions and international students' hindrances: A case of students from the African Portuguese-Speaking countries at two European Portuguese universities. *Journal of International Students, 7*(2), 367-394.

Amirali, S., & Bakken, J. P. (2015). Trends and challenges of recruiting and retaining international graduate students: An internal perspective. *Journal of Education Research*, *9*(4), 425-433.

Astin, A. W. (1984). Student involvement: A developmental theory for higher education. *Journal of College Student Personnel*, *25,* 297-308.

Astin, A. W. (1993). What matters in college: Four critical years revisited. *The Journal of Higher Education*, January 1993. doi: 10.2307/1176821

Astin, A. W. (1999). Student involvement: A developmental theory for higher education. *Journal of College Student Development*, *40*(5), 518-529.

Bandura, A. (1971). Social learning theory. *General Learning Corporation.* Retrieved from http://www.asecib.ase.ro/mps/Bandura_SocialLearningTheory.pdf

Bartlett, J. (2014). The Hosmer-Lemeshow goodness of fit test for logistic regression. *The Stats Geek.* Retrieved from https://thestatsgeek.com/2014/02/16/the-hosmer-lemeshow-goodness-of-fit-test-for-logistic-regression/

Bean, J. P., & Eaton, S. B. (2001). The psychology underlying successful retention practices. *Journal of College Student Retention: Research, Theory & Practice*, *3*(1), 73-89.

Bui, Y. N. (2009). *How to write a master's thesis.* Thousand Oaks, CA: SAGE Publication, Inc.

Cabrera, A. F., Nora, A., Castañeda, M. B. (1993). College persistence: Structural equations modeling test of an integrated model of student retention. *Journal of Higher Education*, *64*(2), 123-139.

Chickering, A. W. (1969). *Education and identity.* San Francisco, CA: Jossey-Bass Inc.

Clark, N. (2009). What defines an international student? A look behind the numbers. *World Education News & Reviews.* Retrieved from https://wenr.wes.org/2009/09/wenr-september-2009-feature

Culatta, R. (2018). Social learning theory (Albert Bandura). *Instructional Design*. Retrieved from http://www.instructionaldesign.org/theories/social-learning/

Curtin, N., Stewart, A. J., & Ostrove, J. M. (2013). Fostering academic self-concept: Advisor support and sense of belonging among international and domestic graduate students. *American Educational Research Journal, 50*(1), 108-137. doi: 10.3102/0002831212446662

Dancer, D., Morrison, K., & Tarr, G. (2015). Measuring the effects of peer learning on students' academic achievement in first-year business statistics. *Studies in Higher Education, 40*(10), 1808-1828. http://dx.doi.org/10.1080/03075079.2014.916671

DeFreitas, S. C. (2012). Differences between African American and European American first-year college students in the relationship between self-efficacy, outcome expectations, and academic achievement. *Social Psychology of Education, 15*(1), 109-123. doi: 10.1007/s11218-011-9172-0

Demetriou, C. & Schmitz-Sciborski, A. (2011). Integration, motivation, strengths and optimism: Retention theories past, present and future. In R. Hayes (Ed.), *Proceedings of the 7th National Symposium on Student Retention, 2011, Charleston*, 300-312.

Edgar, D. W. (2012). Learning theories and historical events affecting instructional design in education: Recitation literacy toward extraction literacy practices. *SAGE Open*, *2*(4), 1-9. doi: 10.1177/2158244012462707

Field, A. (2009). *Discovering statistics using SPSS.* Thousand Oaks, CA: SAGE Publications Inc.

Fischer, K. (2014, June 6). With more international students come retention issues. *The Chronicle of Higher Education,* 323. Retrieved from http://www.universityworldnews.com/article.php?story=20140605182406134&query=international+students

Gay, L. R., Mills, G. E., & Airasian, P. W. (2012). *Educational research: Competencies for analysis and applications.* Upper Saddle River, NJ: Pearson Education, Inc.

Gravetter, F.J., & Wallnau, L.B. (2013). *Statistics for the behavioral sciences*. Belmont, CA: Wadsworth, Cengage Learning.

Heaney, A., & Fisher, R. (2011). Supporting conditionally-admitted students: A case study of assessing persistence in a learning community. *Journal of the Scholarship of Teaching & Learning*, *11*(1), 62-78.

Jama, M. P., Mapesela, M. L. E., & Beylefeld, A. A. (2008). Theoretical perspectives on factors affecting the academic performance of students. *South African Journal of Higher Education, 22*(5), 992-1005. doi: 10.4314/sajhe.v22i5.42919

Kassambara, A. (2017). *Machine learning essentials: Practical guide in R.* Retrieved from http://www.sthda.com/english/articles/36-classification-methods-essentials/148-logistic-regression-assumptions-and-diagnostics-in-r/#logistic-regression-assumptions

Kerlinger, F. N., & Lee, H. B. (2000). *Foundations of behavioral research.* Orlando, FL: Harcourt Inc.

Kupczynski, L., Brown, M., Holland, G. & Uriegas, B. (2014). The relationship between gender and academic success online. *Journal of Educators Online, 11*(1). Retrieved from https://www.learntechlib.org/p/156361/

Lane, D. (n.d). Homoscedasticity. Retrieved from http://davidmlane.com/hyperstat/A121947.html

Lani, J. (2018). Complete dissertation: Assumptions of logistic regression. *Statistics Solutions.* Retrieved from http://www.statisticssolutions.com/assumptions-of-logistic-regression/

Lent, R. W., Miller, M. J., Smith, P. E., Watford, B. A., Lim, R. H., Hui, K., et al. (2013). Social cognitive predictors of adjustment to engineering majors across gender and race/ethnicity. *Journal of Vocational Behavior*, 83, 22-30.

Lund, A., & Lund, M. (2018). Cronbach's alpha (α) using SPSS statistics. Retrieved from https://statistics.laerd.com/spss-tutorials/cronbachs-alpha-using-spss-statistics.php

Mamiseishvili, K. (2012). International student persistence in U.S. postsecondary institutions. *Higher Education, 64*(1), 1-17. Retrieved from https://www.jstor.org/stable/41477916

McGrew, K. S. (2007). Beyond IQ: A model of academic competence and motivation. *Institute for Applied Psychometrics.* Retrieved from http://www.iapsych.com/acmcewok/map.htm

McLeod, S. A. (2016). Bandura - Social learning theory. Retrieved from www.simplypsychology.org/bandura.html

McNeely, J. H. (1938). College student mortality. *Project in Research in Universities*. Washington, DC: United States Government Printing Office.

Mertler, C. A., & Vannatta, R. A. (2013). *Advanced and multivariate statistical methods*. Glendale, CA: Pyrczak.

Milam, J. F. & Berger, J. B. (1997). A modified model of college student persistence: Exploring the relationship between Astin's theory of involvement and Tinto's

theory of student departure. *Journal of College Student Development, 38*(4), 387-400.

Nabavi, R. T. (2012). Bandura's social learning theory & social cognitive learning theory. *Research Gate*. Retrieved from https://www.researchgate.net/publication/267750204

Navarro, R. L., Flores, L. Y., Lee, H. S., Gonzalez, R. (2014). Testing a longitudinal social cognitive model of intended persistence with engineering students across gender and race/ethnicity. *Journal of Vocational Behavior, 85*(1), 146-155.

Newell, E. M. (2015). International student-athlete adjustment issues: Advising recommendations for effective transitions. *NACADA Journal, 35*(2), 36-47.

Osborne, J. W., & Waters, E. (2002). Four assumptions of multiple regression that researchers should always test. *Practical Assessment, Research & Evaluation*, *8*(2). Retrieved from https://pareonline.net/getvn.asp?v=8&n=2

Pandey, R. (2017). What is logistic regression? *Quora.* Retrieved from https://www.quora.com/What-is-logistic-regression

Pascarella, E. (1985). College environmental influences on learning and cognitive development: A critical review and synthesis. In J. Smart (Ed.), Higher education: Vol. 1. Handbook of theory and research (pp. 1-64). New York, NY: Agathon.

Pascarella, E. T., Pierson, C. T., Wolniak, G. C., Terenzini, P. T. (2004). First-generation college students: Additional evidence on college experiences and outcomes. *The Journal of Higher Education*, *75*(3), 249-284. doi: 10.1353/jhe.2004.0016

Pascarella, E. T., Terenzini, P. T. (1991). *How college affects students: Findings and insights from twenty years of research.* San Francisco, CA: Jossey-Bass Inc.

Perez-Encinas, A., & Ammigan, R. (2016). Support services at Spanish and U.S. institutions: A driver for international student satisfaction. *Journal of International Students, 6*(4), 984-998. Retrieved from http://jistudents.org/

Perry, C. J., Lausch, D. W., Weatherford, J., Goeken, R., & Aemendares, M. (2017). International student's perceptions of university life. *College Student Journal*, *51*(2), 279-290.

Rebolledo-Gómez, C., & Ranchin, J. (2013, July 5). Education indicators in focus: How is international student mobility shaping up? *OECD*. Retrieved from http://www.oecd.org/education/skills-beyond-school/EDIF%202013--N%C2%B014%20(eng)-Final.pdf

Sanchez, J. E., & Smith, J. (2017). Non-U.S. citizen, community college students: Their federal student aid status, gender, achievement, and persistence at an emerging HIS. *Journal of Student Financial Aid, 47*(3), Article 3. Retrieved from http://publications.nasfaa.org/jsfa/vol47/iss3/3

Schulmann, P., & Choudaha, R. (2014). International student retention and success: A comparative perspective. *Mobility Monitor*, 12-19.

Second Language Testing Inc. (2018). Pilot testing and field testing. *Second Language Testing Inc.* Retrieved from https://www.2lti.com/test-development/pilot-testing-and-field-testing/

Singaram, V. S., Van Der Vleuten, C. M., Muijtjens, A. M., & Dolmans, D. M. (2012). Relationships between language background, secondary school scores, tutorial group processes, and students' academic achievement in PBL: Testing a causal model. *Interdisciplinary Journal of Problem-Based Learning*, *6*(1), 153-164.

Steinmayr, R., Meißner, A., Weidinger, A. F., & Wirthwein, L. (2014). Academic achievement. *Oxford Bibliographies.* doi: 10.1093/obo/9780199756810-0108

Stewart, S., Lim, D. H., & Kim, J. (2015). Factors influencing college persistence for first-time students. *Journal of Developmental Education, 38*(3). Retrieved from https://files.eric.ed.gov/fulltext/EJ1092649.pdf

Strand, S., Cadwallader, S., & Firth, D. (2011). Using statistical regression methods in education research: Assumptions for multiple regression. *ESRC National Center for Research Methods*. University of Southampton. Retrieved from http://www.restore.ac.uk/srme/www/fac/soc/wie/research-new/srme/modules/mod3/3/index.html

Sukhlecha, A., Jadav, S. P., Gosai, T. R., & Balusamy, D. (2016). Student-led objective tutorials in Pharmacology: An interventional study. *Indian Journal of Pharmacology*, *48*(Suppl 1)*,* S83-S88. doi:10.4103/0253-7613.193310

Sullivan, W. D., Jr. (2018). A case study exploring international student engagement at three small, private colleges. *Journal of International Students, 8*(2), 977-1001. doi: 10.5281/zenodo.1250397

Tinto, V. (2017). Reflections on student persistence. *Student Success, 8(*2), 1-8. doi: 10.5204/ssj.v8i2.376

University of Phoenix (2015). IRB corner: Field testing, pilot studies, and IRB review timing. Retrieved from https://research.phoenix.edu/news/irb-corner-august-2015

Walberg, H. J. (1981). A psychological theory of educational productivity. In F. H. Farley & N. Gordon (Eds.), *Psychology and Education: The state of the union* (pp. 81-108). Berkeley, CA: McCutchan.

Walberg, H. J. (1984). Improving the productivity of America's schools. *Educational Leadership*, *41,* 19-30.

York, T. T., Gibson, C., & Rankin, S. (2015). Defining and measuring academic success. *Practical Assessment, Research & Evaluation, 20*(5). Retrieved from http://pareonline.net/getvn.asp?v=20&n=5

Young, B. N., & Snead, D. (2017). Saudi Arabian international graduate students' lived experiences at a U.S. university. *Journal of Learning in Higher Education, 13*(2), 39-44.

Zhang, Y. L., & Dinh, T. V. (2017). Advising international students in engineering programs: Academic advisors' perceptions of intercultural communication competence. *NACADA Journal*, *37*(2), 33-43. Retrieved from https://tsuhhelweb.tsu.edu:2553/10.12930/NACADA-15-028